

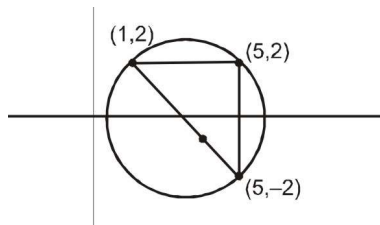
Exercise J01

Q1. The radius of the circle passing through the points $(1, 2)$, $(5, 2)$ & $(5, -2)$ is:

- (A) $5\sqrt{2}$ (B) $2\sqrt{5}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

Ans. D

Sol.



$$\text{diameter} = 4\sqrt{2} \Rightarrow r = 2\sqrt{2}$$

Q2. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area as 154 sq. units. Then the equation of the circle is (Take $\pi = 22/7$)

- (A) $x^2 + y^2 - 2x + 2y = 62$ (B) $x^2 + y^2 + 2x - 2y = 62$ (C) $x^2 + y^2 + 2x - 2y = 47$
 (D) $x^2 + y^2 - 2x + 2y = 47$

Ans. D

Sol. Centre of circle = Point of intersection of diameters = $(1, -1)$

$$\text{Now area} = 154 \Rightarrow \pi r^2 = 154 \Rightarrow r = 7$$

Hence the equation of required circle is

$$(x - 1)^2 + (y + 1)^2 = 7^2 \Rightarrow x^2 + y^2 - 2x + 2y = 47$$

Q3. The equation of the circle passing through the points $(1, 0)$ and $(0, 1)$ and having the smallest radius is :

- (A) $x^2 + y^2 + x + y - 2 = 0$ (B) $x^2 + y^2 - 2x - 2y + 1 = 0$ (C) $x^2 + y^2 - x - y = 0$
 (D) $x^2 + y^2 + 2x + 2y - 7 = 0$

Ans. C

Sol. $(1, 0)$ and $(0, 1)$ will be ends of diameter

So equation of circle

$$(x - 1)(x - 0) + (y - 0)(y - 1) \\ x^2 + y^2 - x - y = 0$$

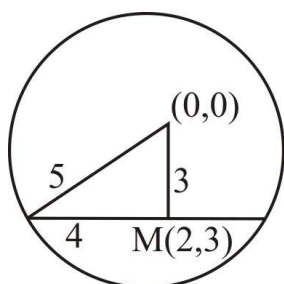
Q4. The two lines through $(2, 3)$ from which the circle $x^2 + y^2 = 25$ intercepts chords of length 8 units have equations

- (A) $2x + 3y = 13$, $x + 5y = 17$ (B) $y = 3$, $12x + 5y = 39$ (C) $x = 2$, $9x - 11y = 51$

(D) none of these

Ans. B

Sol.



Let slope of required line is m

$$y - 3 = m(x - 2) \Rightarrow mx - y + (3 - 2m) = 0$$

length of $\perp$ from origin = 3

$$\Rightarrow 9 + 4m^2 - 12m = 9 + 9m^2 \Rightarrow 5m^2 + 12m = 0 \Rightarrow m = 0, -\frac{12}{5}$$

Hence lines are $y - 3 = 0 \Rightarrow y = 3$

$$y - 3 = -\frac{12}{5}(x - 2) \quad \frac{12}{5} 5y - 15 = -12x + 24 \quad \frac{12}{5} 12x + 5y = 39.$$

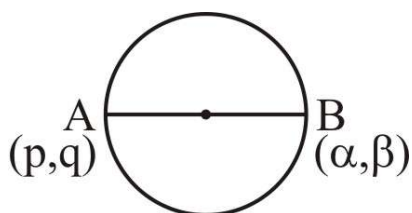
Q5. A variable circle passes through the fixed point A (p, q) and touches x-axis. The locus of the other end of the diameter through A is

(A) $(y-q)^2 = 4px$ (B) $(x-q)^2 = 4py$ (C) $(y-p)^2 = 4qx$ (D) $(x-p)^2 = 4qy$

Ans. D

Sol. Equation of circle having AB as diameter is

$$(x - p)(x - \alpha) + (y - q)(y - \beta) = 0$$



$$\text{or } x^2 + y^2 - (p + \alpha)x - (q + \beta)y + p\alpha + q\beta = 0 \quad \dots (i)$$

as it touches x-axis putting $y = 0$,

$$\text{we get } x^2 - (p + \alpha)x + p\alpha + q\beta = 0 \quad \dots (ii)$$

Since, circle (i) touches x-axis

Discriminant of equation (ii) = 0

$$\Rightarrow (p + \alpha)^2 = 4(p\alpha + q\beta) \Rightarrow (p - \alpha)^2 = 4q\beta$$

$\therefore$ Locus of B(α, β) is $(p - x)^2 = 4qy$

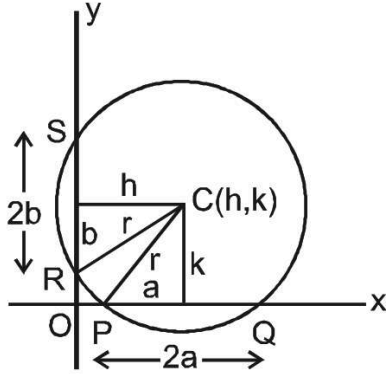
$$\text{or } (x - p)^2 = 4qy$$

Q6. Two thin rods AB & CD of lengths $2a$ & $2b$ move along OX & OY respectively, when 'O' is the origin. The equation of the locus of the centre of the circle passing through the extremities of the two rods is:

- (A) $x^2 + y^2 = a^2 + b^2$ (B) $x^2 - y^2 = a^2 - b^2$ (C) $x^2 + y^2 = a^2 - b^2$
 (D) $x^2 - y^2 = a^2 + b^2$

Ans. B

Sol. $h^2 + b^2 = r^2 \Rightarrow k^2 + a^2 = r^2 \Rightarrow h^2 - k^2 = a^2 - b^2$



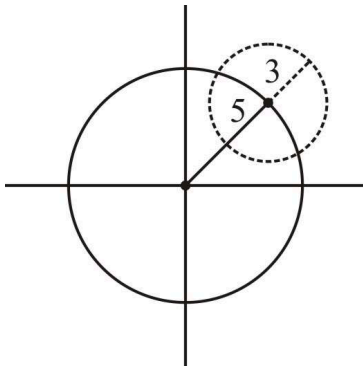
locus is $x^2 - y^2 = a^2 - b^2$

Q7. The centre of a set of circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point in the set is

- (A) $4 \leq x^2 + y^2 \leq 64$ (B) $x^2 + y^2 \leq 25$ (C) $x^2 + y^2 \geq 25$ (D) $3 \leq x^2 + y^2 \leq 9$

Ans. A

Sol.



$4 \leq x^2 + y^2 \leq 64$

Q8. If $\frac{x^2 + y^2}{x + y} = 4$, then all possible values of $(x - y)$ is given by

- (A) $[-2\sqrt{2}, 2\sqrt{2}]$ (B) $\{-4, 4\}$ (C) $[-4, 4]$ (D) $[-2, 2]$

Ans. C

Sol. We have $x^2 + y^2 = 4x + 4y$
 $\Rightarrow (x - 2)^2 + (y - 2)^2 = 8 = (2\sqrt{2})^2$

Hence x, y lie on a circle with parametric coordinates

$$\begin{aligned} \begin{matrix} x=2+2\sqrt{2}\cos\theta \\ y=2+2\sqrt{2}\sin\theta \end{matrix} &\Rightarrow x-y=2\sqrt{2}(\cos\theta-\sin\theta) \\ \therefore x-y &\in [-4, 4] \text{ Ans. } \end{aligned}$$

Q9. The circle $x^2 + y^2 - 6x - 10y + c = 0$ does not intersect or touch either axis, & the point $(1, 4)$ is inside the circle. Then the range of possible values of c is given by:

- (A) $c > 9$ (B) $c > 25$ (C) $c > 29$ (D) $25 < c < 29$

Ans. D

Sol. $g^2 < c$ and $f^2 < c$ and $s, < 0$
 $g < c$, $25 < c$, $c < 29$
 $25 < c < 29$

Q10. The tangent lines to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the line $4x + 3y + 5 = 0$ are given by:

- (A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$ (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$
 (C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$ (D) none of these

Ans. B

Sol. Line parallel to given line $4x + 3y + 5 = 0$ is $4x + 3y + k = 0$
 This is tangent to $x^2 + y^2 - 6x + 4y - 12 = 0$
 $\left| \frac{12 - 6 + k}{5} \right| = 5 \Rightarrow 6 + k = \pm 25 \Rightarrow k = 19, -31$
 Hence required line $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$

Q11. The condition so that the line $(x + g) \cos\theta + (y + f) \sin\theta = k$ is a tangent to $x^2 + y^2 + 2gx + 2fy + c = 0$ is

- (A) $g^2 + f^2 = c + k^2$ (B) $g^2 + f^2 = c^2 + k$ (C) $g^2 + f^2 = c^2 + k^2$ (D) $g^2 + f^2 = c + k$

Ans. A

Sol. $p = \left| \frac{(-g + g)\cos\theta + (-f + f)\sin\theta - k}{\sqrt{\cos^2\theta + \sin^2\theta}} \right| = \sqrt{g^2 + f^2 - c} \Rightarrow g^2 + f^2 = c + k^2$

Q12. The tangent to the circle $x^2 + y^2 = 5$ at the point $(1, -2)$ also touches the circle $x^2 + y^2 - 8x + 6y + 20 = 0$ at

- (A) $(-2, 1)$ (B) $(-3, 0)$ (C) $(-1, -1)$ (D) $(3, -1)$

Ans. D

Sol. Equation of tangent $x - 2y = 5$

Let required point be (α, β)

$$\alpha x + \beta y - 4(x + \alpha) + 3(y + \beta) + 20 = 0 \Rightarrow x(\alpha - 4) + y(\beta + 3) - 4\alpha + 3\beta + 20 = 0$$

Comparing

$$\frac{\alpha - 4}{1} = \frac{\beta + 3}{-2} = \frac{4\alpha - 3\beta - 20}{5}$$

Similarly $(\alpha, \beta) (3, -1)$

- Q13. The lengths of the tangents from any point of the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$, $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio:

(A) 1 : 2 (B) 2 : 3 (C) 3 : 4 (D) 1 : 4

Ans. A

Sol. Let any point $P(x_1, y_1)$ to the circle $x^2 + y^2 - \frac{16x}{5} + \frac{64y}{15} = 0 \Rightarrow x_1^2 + y_1^2 - \frac{16}{5}x_1 + \frac{64}{15}y_1 = 0$

Length of tangent from $P(x_1, y_1)$ to the circle are in ration

$$\frac{\sqrt{S_1}}{\sqrt{S_2}} = \frac{\sqrt{x_1^2 + y_1^2 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}}{\sqrt{x_1^2 + y_1^2 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}} = \sqrt{\frac{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{24}{5}x_1 + \frac{32}{5}y_1 + 15}{\frac{16}{5}x_1 - \frac{64}{15}y_1 - \frac{48}{5}x_1 + \frac{64}{5}y_1 + 60}}$$

$$\sqrt{\frac{-24x_1 + 32y_1 + 225}{-96x_1 + 128y_1 + 900}} = \sqrt{\frac{-24x_1 + 32y_1 + 225}{4(-24x_1 + 32y_1 + 225)}} = \frac{1}{2}$$

- Q14. If L_1 and L_2 are the length of the tangent from $(0, 5)$ to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - y + 1 = 0$ then

(A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$

Ans. C

Sol. [Hint: $L_1 = L_2 = \sqrt{21}$]

- Q15. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is

(A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

Ans. C

Sol. $Q(0, 3)$

$$\text{Length PQ} = \sqrt{S_1}$$

$$= \sqrt{0 + 9 + 0 + 18 - 2} = 5$$

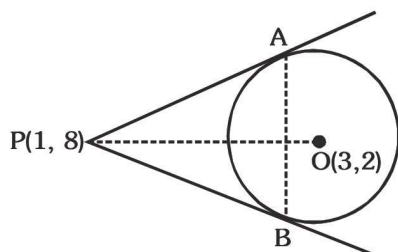
- Q16. Tangents drawn from the point $P(1, 8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is

(A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$

(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$

Ans. B

Sol.



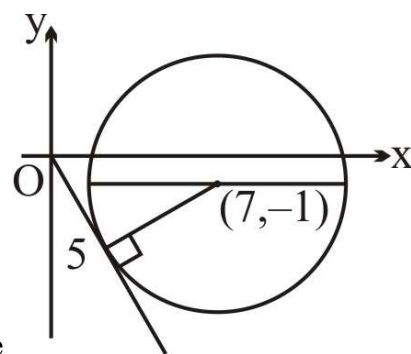
The required circle is a circle described on OP as diameter.

Q17. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. C

Sol.



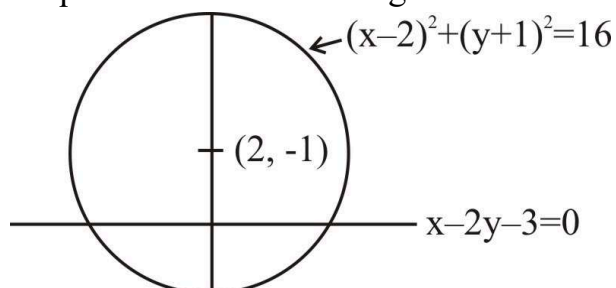
$\left. \begin{array}{l} \text{radius} = 5 \\ \text{length of tangent} = 5 \\ \therefore \text{the figure is square} \end{array} \right\} \text{angle} = 90^\circ$ origin lies on the director circle

Q18. The equation of the diameter of the circle $(x - 2)^2 + (y + 1)^2 = 16$ which bisects the chord cut off by the circle on the line $x - 2y - 3 = 0$ is

(A) $x + 2y = 0$ (B) $2x + y - 3 = 0$ (C) $3x + 2y - 4 = 0$ (D) none

Ans. B

Sol. Required diameter is $\perp$ to given line.



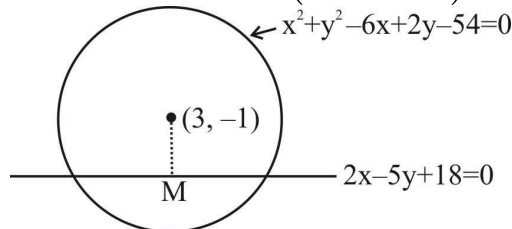
Hence $y + 1 = -2(x - 2) \Rightarrow 2x + y - 3 = 0$

- Q19. The co-ordinates of the middle point of the chord cut off on $2x - 5y + 18 = 0$ by the circle $x^2 + y^2 - 6x + 2y - 54 = 0$ are
 (A) (1, 4) (B) (2, 4) (C) (4, 1) (D) (1, 1)

Ans. A

Sol. Required point is foot of $\perp$

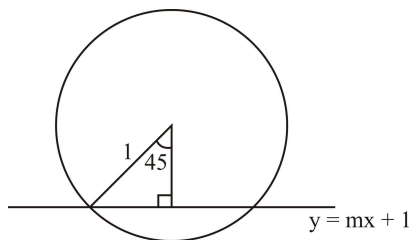
$$\frac{x-3}{2} = \frac{y+1}{-5} = -\left(\frac{6+5+18}{4+25}\right) = -1 \Rightarrow x = 1, y = 4$$



- Q20. If the chord $y = mx + 1$ of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then value of m is
 (A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$ (C) ± 1 (D) None of these

Ans. C

Sol.



$$\left| \frac{1}{1+m^2} \right| = \frac{1}{\sqrt{2}}$$

$$m = \pm 1$$

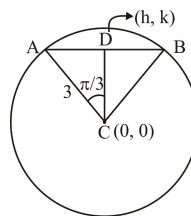
- Q21. Let C be the circle with centre $(0, 0)$ and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre is

(A) $x^2 + y^2 = \frac{3}{2}$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = \frac{27}{4}$ (D) $x^2 + y^2 = \frac{9}{4}$

Ans. D

Sol. Let AB be the chord subtending angle $\frac{2\pi}{3}$ at the centre C of circle
 Now, $\angle ACD = \frac{\pi}{3}$

Let the coordinates of midpoint D be (h, k)



$$\text{In } \triangle ACD, \cos = \frac{\pi}{3} = \frac{CD}{CA}$$

$$\Rightarrow \frac{1}{2} = \frac{\sqrt{h^2 + k^2}}{3}$$

$$\Rightarrow x^2 + y^2 = \frac{9}{4}, \text{ which is the required locus.}$$

Q22. The chord of contact of the tangents drawn from a point on the circle, $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$ then a, b, c are in :

(A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

Ans. B

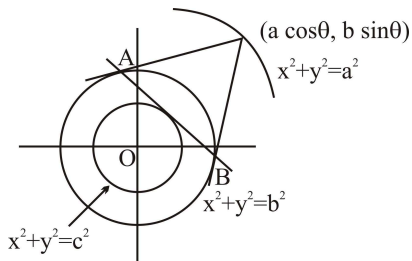
Sol. equation of chord of contact AB

$$x a \cos \theta + y a \sin \theta = b^2 \quad \dots (1)$$

this is tangent to the circle $x^2 + y^2 = c^2$

$\therefore$ perpendicular from (0, 0) on (1) is equal to c

$$\Rightarrow \frac{b^2}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} = c \Rightarrow b^2 = ac$$



Q23. If tangent at (1, 2) to the circle $c_1 : x^2 + y^2 = 5$ intersects the circle $c_2 : x^2 + y^2 = 9$ at A and B and tangents at A and B to the second circle meet at point C, then the co-ordinates of C are :

(A) (4, 5) (B) (9/15, 18/5) (C) (4, -5) (D) (9/5, 18/5)

Ans. D

Sol. Let 'c' be (x_1, y_1) . It's chord of contact w.r.t. c_2 is $x x_1 + y y_1 = 9$. Also the tangent at (1, 2) to c_1 is $x + 2y = 5$.

Now compare to get x_1, y_1

Q24. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is

- (A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$ (C) $9(x^2 + y^2) = (2x + 3y)^2$
 (D) $x^2 + y^2 = (2x + 3y)^2$

Ans. C

Sol. one of the tangent is y - axis i.e. $x = 0$ and only (C) satisfies $x = 0$

Q25. If $x = 3$ is the chord of contact of the circle $x^2 + y^2 = 81$, then the equation of the corresponding pair of tangents, is

- (A) $x^2 - 8y^2 + 54x + 729 = 0$ (B) $x^2 - 8y^2 - 54x + 729 = 0$
 (C) $x^2 - 8y^2 - 54x - 729 = 0$ (D) $x^2 - 8y^2 = 729$

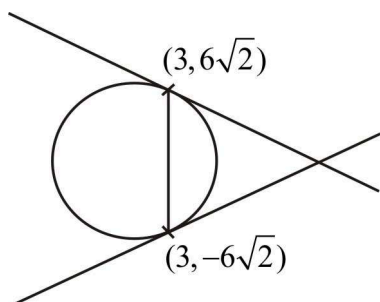
Ans. B

Sol. Put $x = 3$ in the equation of circle to get the point of contact, $y = \pm 6\sqrt{2}$

Hence, equation of the pair of tangents is

$$(3x + 6\sqrt{2}y - 81)(3x - 6\sqrt{2}y - 81) = 0 \Rightarrow (x - 27)^2 - 8y^2 = 0$$

$$x^2 - 54x - 8y^2 + 729 = 0$$



Q26. Intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$, is AB. Equation of the circle on AB as a diameter is

- (A) $x^2 + y^2 + x - y = 0$ (B) $x^2 + y^2 - x + y = 0$ (C) $x^2 + y^2 + x + y = 0$
 (D) $x^2 + y^2 - x - y = 0$

Ans. D

Sol. Given, circle is $x^2 + y^2 - 2x = 0$ (i)

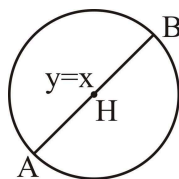
and line is $y = x$ (ii)

Putting $y = x$ in (i),

We get $2x^2 - 2x = 0 \Rightarrow x = 0, 1$

From (i), $y = 0, 1$

Let $A = (0, 0)$, $B = (1, 1)$



Equation of required circle is

$$(x - 0)(x - 1) + (y - 0)(y - 1) = 0$$

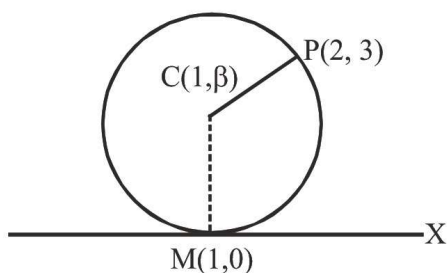
$$\text{or } x^2 + y^2 - x - y = 0$$

Q27. The length of the diameter of the circle which touches the x-axis at the point (1, 0) and passes through the point (2, 3) is :

- (A) $\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $\frac{3}{5}$ (D) $\frac{6}{5}$

Ans. B

Sol.



Let center of the circle be $C(1, \beta)$

$$\beta^2 = (2 - 1)^2 + (3 - \beta)^2$$

$$\Rightarrow \beta^2 = -6\beta + 10 + \beta^2$$

$$\Rightarrow \beta = \frac{5}{3}$$

$$\therefore r = \frac{5}{3}$$

$$\text{diameter} = \frac{10}{3}$$

Q28. The circle passing through the point $(-1, 0)$ and touching the y-axis at $(0, 2)$ also passes through the point :

- (A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 0\right)$ (C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

Ans. D

Sol. Family of circle which touches y-axis at $(0, 2)$ is

$$x^2 + (y - 2)^2 + \lambda x = 0$$

Passing through $(-1, 0)$

$$\Rightarrow 1 + 4 - \lambda = 0 \Rightarrow \lambda = 5$$

$$\therefore x^2 + y^2 + 5x - 4y + 4 = 0$$

which satisfy the point $(-4, 0)$

Q29. A circle touches the x-axis and also touches the circle with centre at $(0, 3)$ and radius 2. The locus of the centre of the circle is

- (A) an ellipse (B) a circle (C) a hyperbola (D) a parabola

Ans. D

Sol. Let centre $^o(h, k)$; As $C_1C_2 = r_1 + r_2$, (Given)

$$\Rightarrow \sqrt{(h-0)^2 + (k-3)^2} = |k+2|$$

$$\Rightarrow h^2 = 5(2k-1)$$

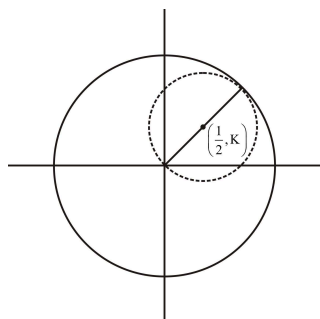
Hence locus, $x^2 = 5(2y-1)$, which is parabola

Q30. The centre of the circle passing through (0,0) and (1, 0) and touching the circle $x^2+y^2=9$ is

- (A) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, -\sqrt{2}\right)$ (C) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right)$

Ans. B

Sol.



$$C_1C_2 = \frac{r}{2}$$

$$\frac{1}{4} + k^2 = \left(\frac{3}{2}\right)^2$$

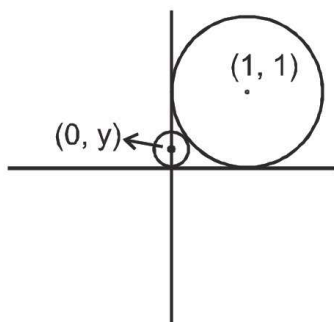
$$k = \pm\sqrt{2}$$

Q31. Let C be the circle with centre at (1, 1) and radius = 1. If T is the circle centred at (0, y), passing through origin and touching the circle C externally, then the radius of T is equal to :

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Ans. B

Sol.



$$c_1 (1, 1) \quad r_1 = 1$$

$$c_2 (0, y) \quad r_2 = |y|$$

$$c_1 c_2 = r_1 + r_2$$

$$\sqrt{(1-0)^2 + (1-y)^2} = 1 + |y|$$

$$2 - 2y + y^2 = y^2 + 2|y| + 1$$

$$4|y| = 1$$

$$|y| = \frac{1}{4}$$

$$y = \frac{1}{4}$$

Q32. The equation of the common tangent to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$ at their point of contact is

(A) $12x + 5y + 19 = 0$ (B) $5x + 12y + 19 = 0$ (C) $5x - 12y + 19 = 0$

(D) $12x - 5y + 19 = 0$

Ans. B

Sol. Equation of common tangent at point of contact is $S_1 - S_2 = 0$

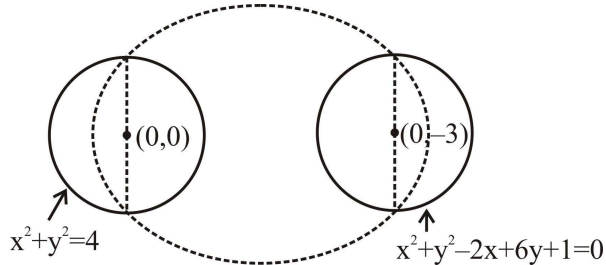
$$\Rightarrow 10x + 24y + 38 = 0 \Rightarrow 5x + 12y + 19 = 0$$

Q33. The locus of the centre of the circle which bisects the circumferences of the circles $x^2 + y^2 = 4$ & $x^2 + y^2 - 2x + 6y + 1 = 0$ is:

- (A) a straight line (B) a circle (C) a parabola (D) pair of straight line

Ans. A

Sol. Let required circle is $x^2 + y^2 + 2gx + 2fy + c = 0$



Hence common chord with $x^2 + y^2 - 4 = 0$
is $2gx + 2fy + c + y = 0$

This is diameter of circle $x^2 + y^2 = 4$ hence $c = -4$.

Now again common chord with other circle

$$2x(g+1) + 2y(f-3) + (c-1) = 0$$

This is diameter of $x^2 + y^2 - 2x + 6y + 1 = 0$

$$2(g+1) - 6(f-3) + 5 = 0$$

$$2g - 6f + 15 = 0$$

locus $2x - 3y - 15 = 0$ which is st. line.

Q34. The circumference of the circle $x^2 + y^2 - 2x + 8y - q = 0$ is bisected by the circle $x^2 + y^2 + 4x + 12y + p = 0$, then $p + q$ is equal to:

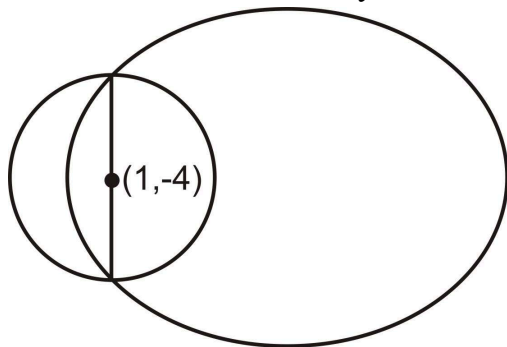
- (A) 25 (B) 100 (C) 10 (D) 48

Ans. C

Sol. Common chord of given circle

$$6x + 4y + (p + q) = 0$$

This is diameter of $x^2 + y^2 - 2x + 8y - q = 0$



centre $(1, -4)$

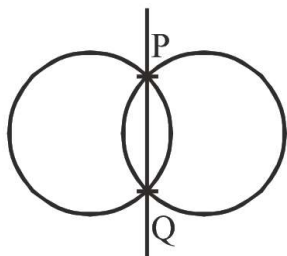
$$6 - 16 + (p + q) = 0 \Rightarrow p + q = 10$$

Q35. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and (1, 1) for :

- (A) all except one value of p (B) all except two values of p
 (C) exactly one value of p (D) all values of p

Ans. A

Sol.



Eqn. of line PQ

$$x + 5y + 2p - 5 + p^2 = 0$$

P, Q and (1, 1)

will not lie on a circle of (1, 1)

Lies on the line

$$x + 5y + p^2 + 2p - 5 = 0$$

$$\Rightarrow 1 + 5 + p^2 + 2p - 5 = 0$$

$$p^2 + 2p + 1 = 0$$

$$\Rightarrow p = -1$$

Therefore there is a circle passing through P, Q and (1, 1) for all values of p.

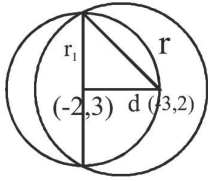
Except $p = -1$.

Q36. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at $(-3, 2)$, then the radius of S is :

- (A) 5 (B) 10 (C) $5\sqrt{2}$ (D) $5\sqrt{3}$

Ans. D

Sol.



$$r_1 = 5$$

$$d = \sqrt{50}$$

$$r = \sqrt{r_1^2 + d^2}$$

$$= \sqrt{75} = 5\sqrt{3}$$

Q37. If the two circles, $x^2 + y^2 + 2g_1x + 2f_1y = 0$ & $x^2 + y^2 + 2g_2x + 2f_2y = 0$ touch each other then:

- (A) $f_1 g_1 = f_2 g_2$ (B) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$ (C) $f_1 f_2 = g_1 g_2$ (D) Done

Ans. B

Sol. If two circles touch each other, then

$$C_1 C_2 = r_1 + r_2$$

$$\sqrt{(-g_1 + g_2)^2 + (-f_1 + f_2)^2} = \sqrt{g_1^2 + f_1^2} + \sqrt{g_2^2 + f_2^2} \text{ squaring both sides}$$

$$-2g_1g_2 - 2f_1f_2 = 2\sqrt{(g_1^2 + f_1^2)(g_2^2 + f_2^2)} \Rightarrow (g_1f_2)^2 - 2g_1g_2f_1f_2 = 0 \Rightarrow \frac{g_1}{g_2} = \frac{f_1}{f_2}$$

Q38. Equation of the circle cutting orthogonally the three circles $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$ is

- (A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$
 (C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) none of these

Ans. A

Sol. $S_1 - S_2 = 0 \Rightarrow 7x - 8y + 16 = 0$
 $S_2 - S_3 = 0 \Rightarrow 2x - 4y + 20 = 0$
 $S_3 - S_1 = 0 \Rightarrow 9x - 12y + 36 = 0$

On solving centre (8, 9)

Length of tangent $\sqrt{S_1} = \sqrt{64 + 81 - 16 + 27 - 7} = \sqrt{149} = (x - 8)^2 + (y - 9)^2 = 149$
 $= x^2 + y^2 - 16x - 18y - 4 = 0$

Q39. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the locus of its centre is

- (A) $2ax - 2by - (a^2 + b^2 + 4) = 0$ (B) $2ax + 2by - (a^2 + b^2 + 4) = 0$
 (C) $2ax - 2by + (a^2 + b^2 + 4) = 0$ (D) $2ax + 2by + (a^2 + b^2 + 4) = 0$

Ans. B

Sol. Let the variable circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (i)$$

Circle (i) cuts circle $x^2 + y^2 - 4 = 0$ orthogonally

$$\Rightarrow 2g \cdot 0 + 2f \cdot 0 = c - 4 \Rightarrow c = 4$$

Since circle (i) passes through (a, b)

$$\therefore a^2 + b^2 + 2ga + 2fb + 4 = 0$$

$\therefore$ Locus of centre $(-g, -f)$ is

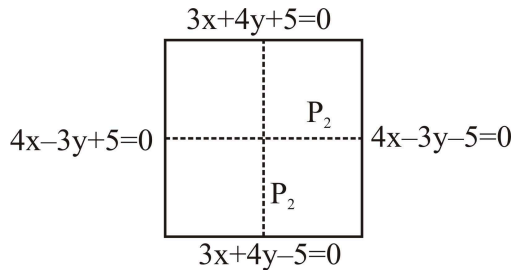
$$2ax + 2by - (a^2 + b^2 + 4) = 0$$

Exercise J02

Q1. Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.

Ans. square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$

Sol.



$$P_1 = \left(\frac{10}{\sqrt{3^2 + 4^2}} \right) = 2$$

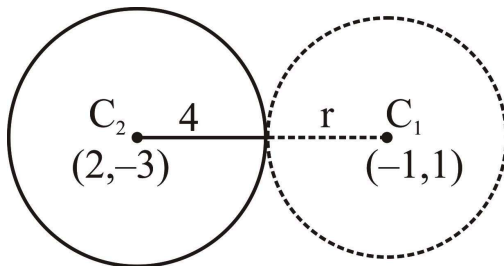
$$P_2 = \left(\frac{10}{\sqrt{3^2 + 4^2}} \right) = 2$$

$$P_1 = P_2 \Rightarrow \text{square}$$

Q2. A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.

Ans. zero, zero

Sol.



$$C_1 C_2 = r + 4$$

equation of circle

$$5 = r + 4$$

$$(x + 1)^2 + (y - 2)^2 = 1$$

$$r = 1$$

$$x^2 + y^2 + 2x - 2y + 1 = 0$$

$$x - \text{intercept} = 0$$

$$y - \text{intercept} = 0$$

- Q3. The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.

Ans. -

Sol. $x = -\frac{my + n}{l} \quad y = -\frac{ln + n}{m}$

Quadratic in x

$$ax^2 + 2hx \left(-\frac{lx + n}{m} \right) + b \left(-\frac{lx + n}{m} \right)^2 - 1 = 0$$

$$ax^2 = \frac{2hlx^2}{m} - \frac{2hx}{m}x + \frac{bl^2x^2}{m^2} + \frac{2blnx}{m^2} + \frac{bn^2}{m^2} - 1 = 0$$

$$\left(a - \frac{2hl}{m} + \frac{bl^2}{m^2} \right) x^2 - \left(\frac{2hx}{m} - \frac{2bln}{m^2} \right) x + \left(\frac{bn^2 - m^2}{m^2} \right) = 0$$

Quadratic y

$$a \left(\frac{my + n}{l} \right)^2 + 2hy \left(-\frac{my + n}{l} \right) + by^2 - 1 = 0$$

$$\frac{am^2}{l^2} y^2 + \frac{2amny}{y^2} + \frac{an^2}{l^2} y^2 = \frac{2hny}{l} + by^2 - 1 = 0$$

$$\left(\frac{am^2}{l^2} + b - \frac{2hm}{l} \right) y^2 + \left(\frac{2amn}{l^2} \right) y + \frac{an^2 - l^2}{l^2} = 0$$

add (1) and (2) and put (0, 0)

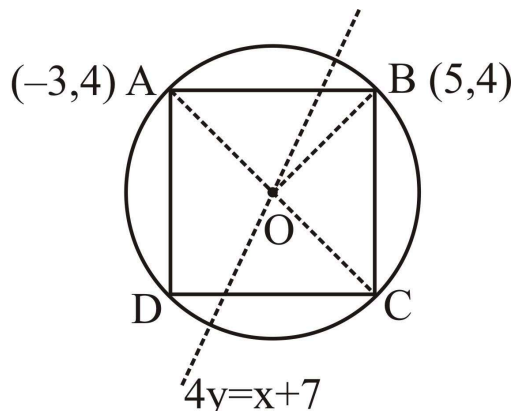
$$\frac{\frac{bn^2 - m^2}{m^2}}{a - \frac{2hl}{m} + \frac{bl^2}{m^2}} + \frac{\frac{an^2 - l^2}{l^2}}{\frac{am^2}{l^2} + b - \frac{2hm}{l}} = 0$$

$$n^2(a + b) = l^2 + m^2$$

- Q4. One of the diameters of the circle circumscribing the rectangle ABCD in order is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle.

Ans. 32 sq. unit

Sol.



$$O\left(h, \frac{h+7}{4}\right)$$

$$OA = OB$$

$$(h+3)^2 + \left(\frac{h+7}{4} - 4\right)^2 = (h-5)^2 + \left(\frac{h+7}{4} - 4\right)^2$$

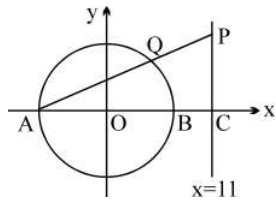
$$h = 1$$

$$O(1, 2)$$

$$AC = 4\sqrt{5}, BC = 4, AB = 8$$

$$\text{Area} = 32$$

- Q5. In the given figure, the circle $x^2 + y^2 = 25$ intersects the x-axis at the point A and B. The line $x = 11$ intersects the x-axis at the point C. Point P moves along the line $x = 11$ above the x-axis and AP intersects the circle at Q. Find



- The coordinates of the point P if the triangle AQB has the maximum area.
- The coordinates of the point P if Q is the middle point of AP.
- The coordinates of P if the area of the triangle AQB is $(1/4)$ th of the area of the triangle APC.

Ans. (i) (11, 16), (ii) (11, 8), (iii) (11, 12)

Sol. (i) $Q(5\cos\theta, 5\sin\theta)$

$$\text{Area} = \left| \frac{1}{2} \times 10 \times (5\sin\theta) \right|$$

$$\text{Area} = 25|\sin\theta|$$

$$(\text{Area})_{\max} \text{ when } \theta = \frac{\pi}{2}$$

$$Q(0, 5)$$

equation of AP

$$x - y + 5 = 0$$

$$\text{put } x = 11$$

$$y = 16$$

$$P(11, 16)$$

$$(ii) A(-5, 0)$$

$$P(11, K)$$

$$Q\left(3, \frac{K}{2}\right)$$

Q lies on circle

$$9 + \frac{K^2}{4} = 25$$

get K

$$(iii) A(-5, 0)$$

$$Q(5\cos\theta, 5\sin\theta)$$

Equation of AP

$$y = \frac{5\sin\theta - 0}{5\cos\theta + 5}(x + 5)$$

$$\text{put } x = 11$$

$$y = \frac{16\sin\theta}{\cos\theta + 1}$$

$$4(\text{Area of } \triangle AQB) = \text{Area of } \triangle APC$$

$$4 \times \frac{1}{2} \times 10 \times 5\sin\theta = \frac{1}{2} \times 16 \times \frac{16\sin\theta}{\cos\theta + 1}$$

$$\cos\theta = \frac{7}{25}$$

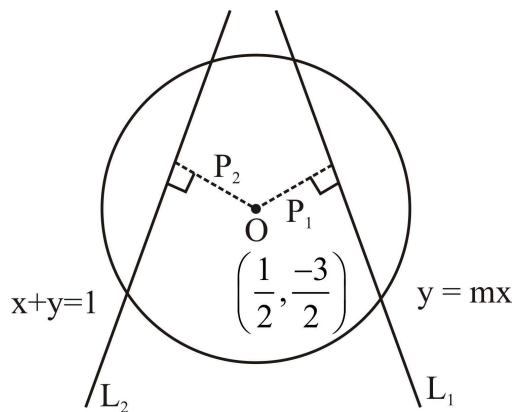
$$y = \frac{16 \times \frac{24}{25}}{\frac{7+25}{25}} = 12$$

$$P(11, 12)$$

Q6. Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1 .

Ans. $x - y = 0$; $x + 7y = 0$

Sol.



$$P_1 = P_2$$

$$\left| \frac{\frac{1}{2}, -\frac{3}{2} - 1}{\sqrt{2}} \right| = \left| \frac{\frac{m}{2} + \frac{3}{2}}{\sqrt{1+m^2}} \right|$$

$$2\sqrt{2} = \left| \frac{m+3}{\sqrt{1+m^2}} \right|$$

$$8 + 8m^2 = m^2 + 6m + 9$$

$$7m^2 - 6m - 1 = 0$$

$$7m^2 - 7m + m - 1 = 0$$

$$m = 1 \quad \text{or} \quad m = \frac{1}{7}$$

Q7. Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.

Ans. $2x - 2y - 3 = 0$

Sol. $\sqrt{r_1^2 - P_1^2} = \sqrt{r_2^2 - P_2^2}$ Let equation of line

$$y = x + C$$

$$4 - \left| \frac{C}{\sqrt{2}} \right|^2 = 9 - \left| \frac{5 - 7 + C}{\sqrt{2}} \right|^2$$

$$4 - \frac{C^2}{2} = 9 - \frac{C^2 - 4C + 4}{2}$$

$$8 - C^2 = 18 - C^2 + 4C - 4$$

$$4C = -6$$

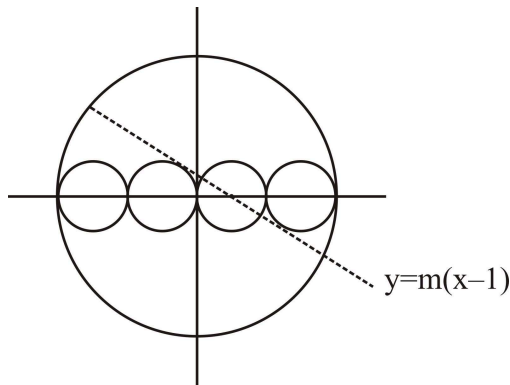
$$C = -\frac{3}{2}$$

$$y = x - \frac{3}{2}$$

- Q8. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

Ans. 63

Sol.



Apply C · O · T w. r. t.

$$\left| \frac{m(-1-1)-0}{\sqrt{1+m^2}} \right| = 1$$

$$m = \pm \frac{1}{\sqrt{3}}$$

$$y = \frac{-1}{\sqrt{3}}x + \frac{1}{\sqrt{3}} \quad \text{or} \quad y = \frac{1}{\sqrt{3}}x + \frac{-1}{\sqrt{3}}$$

$$\text{length} = 2\sqrt{r^2 - p^2}$$

$$= 2\sqrt{16 - \left| \frac{-\frac{1}{\sqrt{3}}}{1^2 + \frac{1}{3}} \right|^2}$$

$$= \sqrt{63}$$

- Q9. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are

M and m respectively, then find the value $(2M + 6m)$.

Ans. 4

Sol. $y - 4 = Z(x - 7)$

Tangent on circle from $(7, 4)$

Z is slope of tangent

$$y = mx \pm \sqrt{m^2 + 1}$$

$$-7 + 4 = \sqrt{Z^2 + 1}$$

$$(4Z-3)(12Z-5)=0$$

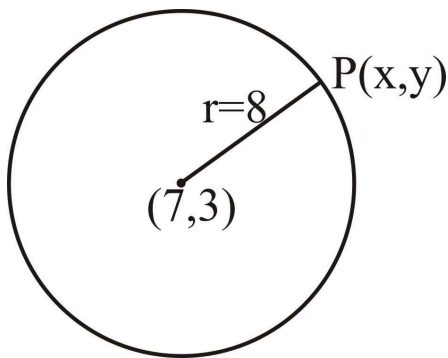
$$Z = \frac{3}{4} \quad \text{or} \quad Z = \frac{5}{12}$$

$$M = \frac{3}{4} \quad \text{and} \quad m = \frac{5}{12}$$

Q10. Given that $x^2 + y^2 = 14x + 6y + 6$, find the largest possible value of the expression $E = 3x + 4y$.

Ans. 73

Sol. $(x-7)^2 + (y-3)^2 = 64 = 8^2$, a circle with centre $(7, 3)$ and radius 8



$$\text{now } \frac{x-7}{\cos\theta} = \frac{y-3}{\sin\theta} = 8$$

$\begin{matrix} x=8\cos\theta+7 \\ y=8\sin\theta+3 \end{matrix}$ lies on the circle for some θ

$$\therefore E = 3x + 4y = 3(8\cos\theta + 7) + 4(8\sin\theta + 3)$$

$$E = (21 + 12) + 24\cos\theta + 32\sin\theta$$

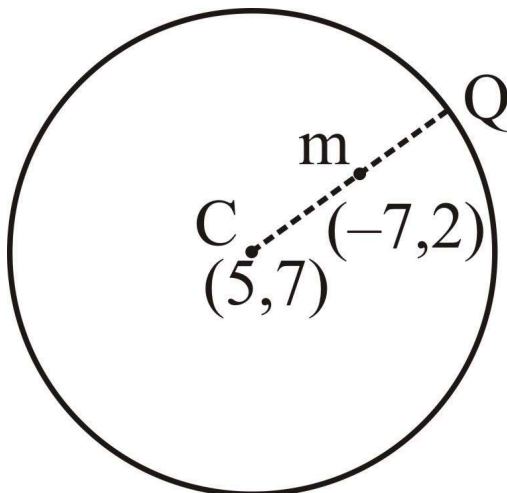
$$= 33 + 8(3\cos\theta + 4\sin\theta)$$

$$\therefore E_{\max} = 33 + 40 = 73 \text{ Ans.}$$

Q11. Find the shortest distance from the point $M(-7, 2)$ to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$.

Ans. 2

Sol.



$$\text{distance} = CQ - Cm = 4 - Cm$$

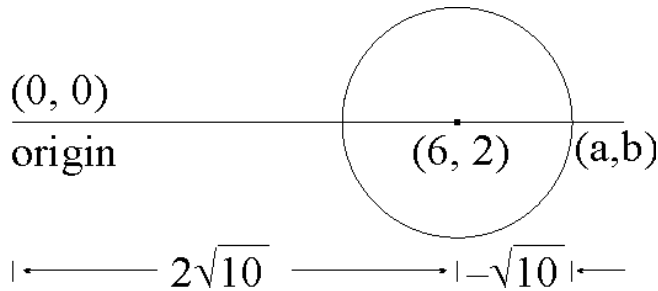
$$= 15 - 13 = 2$$

Q12. Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.

Ans. 9, 3

Sol. radius = $36 + 4 - 30 = \sqrt{10}$

Use section formula



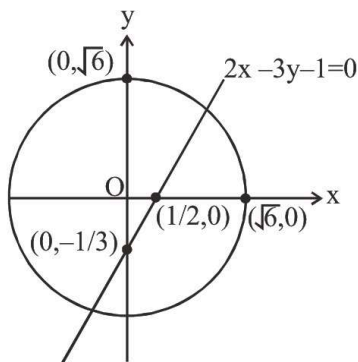
Q13. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If

$$S = \left[\left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right],$$

then the number of point(s) in S lying inside the smaller part is :

Ans. 2

Sol.



If the point lies inside the smaller part, then origin and point should give opposite signs w.r.t. line & point should lie inside the circle.

for origin : $2 \times 0 - 3 \times 0 - 1 = -1$ (-ve)

for $\left(2, \frac{3}{4} \right)$: $2 \times 2 - 3 \times \frac{3}{4} - 1 = \frac{3}{4}$ (+ve); point lies inside the circle

for $\left(\frac{5}{2}, \frac{3}{4} \right)$: $2 \times \frac{5}{2} - 3 \times \frac{3}{4} - 1 = \frac{7}{4}$ (+ve); point lies outside the circle

for $\left(\frac{1}{4}, -\frac{1}{4} \right)$: $2 \times \frac{1}{4} - 3 \times \left(-\frac{1}{4} \right) - 1 = \frac{1}{4}$ (+ve); point lies inside the circle

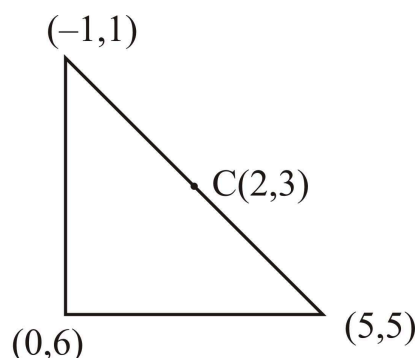
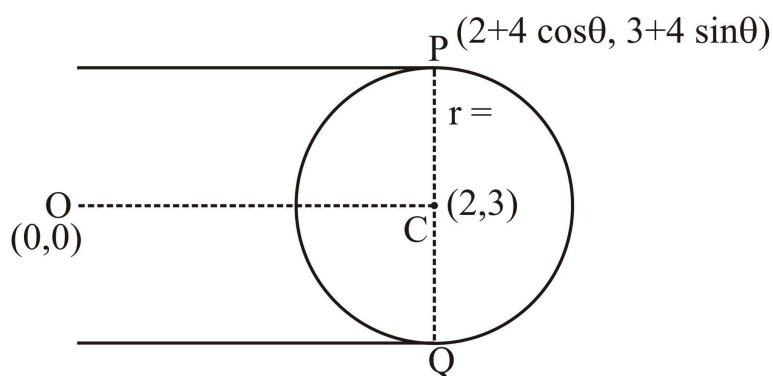
for $\left(\frac{1}{8}, \frac{1}{4} \right)$: $2 \times \frac{1}{8} - 3 \times \left(\frac{1}{4} \right) - 1 = -\frac{3}{2}$ (-ve); point lies inside the circle.

$\therefore$ 2 points lie inside smaller part.

Q14. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.

Ans. (5,1)&(-1,5)

Sol.



$$M_{OC} = \frac{3}{2}$$

$$M_{CP} = \frac{-2}{3}$$

$$\tan \theta = \frac{-2}{3}$$

$\theta \rightarrow \text{II}^{\text{nd}}$ quadrant or IV^{th} quadrant

$$\sin \theta = \frac{2}{\sqrt{13}}$$

$$\sin \theta = \frac{-2}{\sqrt{13}}$$

$$\cos \theta = \frac{3}{\sqrt{13}}$$

$$\cos \theta = \frac{3}{\sqrt{13}}$$

get P and Q

Q15. A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point $(0, 1)$. Find its equation.

Ans. $x^2 + y^2 - 2x - 2y + 1 = 0$ or $x^2 + y^2 - 42x + 38y - 39 = 0$

Sol. centre $(\lambda, 2 - \lambda)$

$$\sqrt{(\lambda - 0)^2 + (2 - \lambda - 1)^2} = \left| \frac{4\lambda - 3(2 - \lambda) + 4}{\sqrt{16 + 9}} \right|$$

$$\sqrt{\lambda^2 + \lambda^2 - 2\lambda + 1} = \left| \frac{7\lambda - 2}{5} \right|$$

$$2\lambda^2 - 2\lambda + 1 = \frac{49\lambda^2 - 28\lambda + 4}{25}$$

$$\lambda^2 - 22\lambda + 21 = 0$$

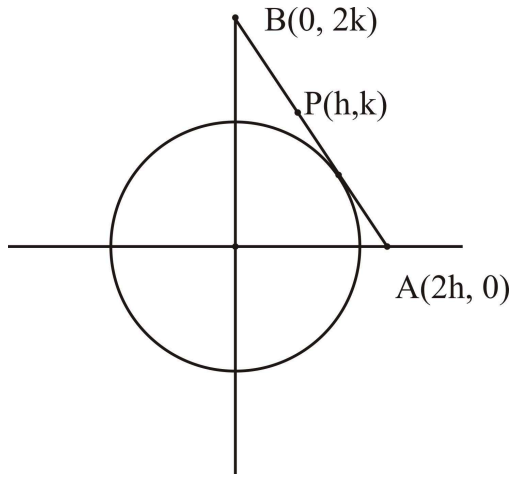
$$\lambda = 21 \quad \text{or} \quad \lambda = 1$$

Centre $(21, -19)$ or $(1, 1)$

Q16. Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

Ans. $a^2(x^2 + y^2) = 4x^2y^2$

Sol.



equation of tangent

$$\frac{x}{2h} + \frac{y}{2k} = 1$$

apply C·O·T

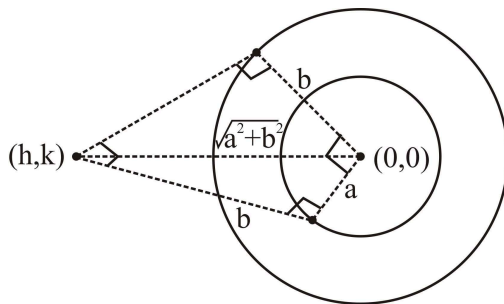
$$\left| \frac{1}{\sqrt{\frac{1}{4h^2} + \frac{1}{4k^2}}} \right| = a$$

$$\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{a}$$

- Q17. Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.

Ans. $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$

Sol.



$$\sqrt{(h-0)^2 + (k-0)^2} = \sqrt{a^2 + b^2}$$

$$x^2 + y^2 = a^2 + b^2$$

- Q18. If the variable line $3x - 4y + k = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, then the range of k is (a, b) where $a, b \in \mathbb{I}$. Find the value of $(b - a)$.

Ans. 6

Sol. $P_1 > r_1$ and $P_2 > r_2$ and $L(C_1) L(C_2) < 0$

Centres lies opposite side of the line

$$P_1 < r_1$$

$$P_2 > r_2$$

$$L(C_1) L(C_2) < 0$$

$$\left| \frac{3-4+k}{5} \right| > 1$$

$$\left| \frac{24-4+k}{5} \right| > 2$$

$$L(1,1) L(8,1) < 0$$

$$|k-1| > 5$$

$$|k+20| > 10$$

$$(3-4+k)(24-4+k) < 0$$

$$(k-1)(k+20) < 0$$

$$k < -4 \text{ or } k > 6 \quad k > -10 \text{ or } k < -30$$

$$-20 < k < 1$$

$$k \in (-10, -4)$$

$$a = -10 \text{ and } b = -4$$

$$-4 + 10 = 6$$

- Q19. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y-axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$, find the value of $a + b$.

Ans. 10

Sol. tangents are parallel

$$2r = \left| \frac{10+6}{\sqrt{2}} \right| = 8\sqrt{2}$$

$$r = 4\sqrt{2} = h$$

$$P_1 = P_2 = h = r$$

$$\left| \frac{h-k+10}{\sqrt{2}} \right| = \left| \frac{h-k-6}{\sqrt{2}} \right| = \sqrt{2}$$

$$(4\sqrt{2} - k - 6)^2 = 64$$

$$(4\sqrt{2} - k - 6)^2 = (8)^2 = 0$$

$$(4\sqrt{2} - k - 6 - 8)(4\sqrt{2} - k - 6 + 8) = 0$$

$$k = 4\sqrt{2} - 14 \text{ or } k = 2 + 4\sqrt{2}$$

$k \rightarrow$ negative

$$\text{Reject } h + k = 2 + 8\sqrt{2}$$

$$a + b = 10$$

- Q20. A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point D $(3, -3)$. Find the following.

(i) Equation of the tangents at A and B.

(ii) Coordinates of the points A and B.

(iii) Angle ADB and the maximum and minimum distances of the point D from the circle.

(iv) Area of quadrilateral ADBC and the ΔDAB .

(v) Equation of the circle circumscribing the ΔDAB and also the intercepts made by this circle on the coordinate axes.

Ans. (i) $3x - 4y = 21$; $4x + 3y = 3$; (ii) A(0, 1) and B (-1, -6); (iii) 90° , $5(\sqrt{2} \pm 1)$ units
(iv) 25 sq. units, 12.5 sq. units; (v) $x^2 + y^2 + x + 5y - 6 = 0$, x intercept 5; y intercept 7

Sol. equation of tangent

$$y + 3 = m(x - 3)$$

C.O.T

$$\left| \frac{-4m - 3m + 2 - 3}{\sqrt{1 + m^2}} \right| = 5$$

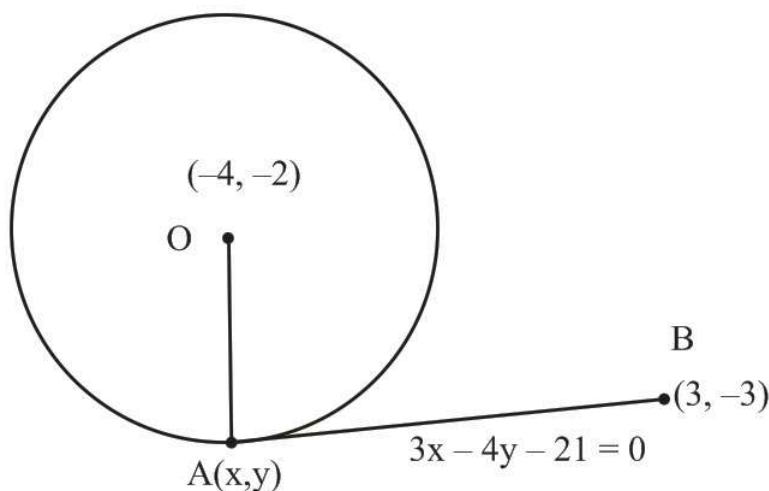
$$49m^2 + 14m - 24 = 0$$

$$12m^2 + 7m - 12 = 0$$

$$4m(3m + 4) - 3(3m + 4) = 0$$

$$m = \frac{3}{4} \text{ or } m = \frac{-4}{3}$$

$$(i) y + 3 = \frac{3}{4}(x - 3) \text{ and } y + 3 = \frac{-4}{3}(x - 3)$$



equation of OA

$$y + 2 = \frac{-4}{3}(x + 4)$$

$$3y + 6 = -4x - 16$$

$$4x + 3y + 22 = 0$$

A is P.O.I. of $3x - 4y - 21 = 0$ and $4x + 3y + 22 = 0$

similarly get B

$$(iii) m_1 = \frac{3}{4} \text{ and } m_2 = \frac{-4}{3}$$

$$m_1 m_2 = -1$$

$$\text{angle ADB} = \frac{\pi}{2}$$

(iv) ADBC is a square

(v) CB is diameter and mid point of CB is centre of Req. circle.

Q21. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points

$A_1(B_1), A_2(B_2), A_3(B_3), A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of

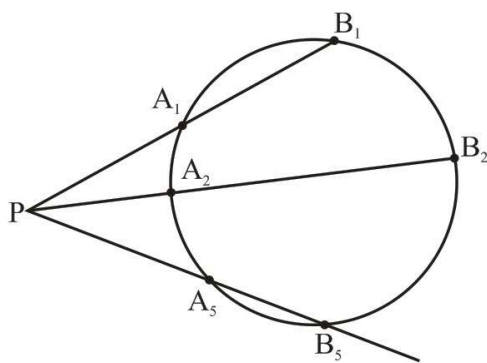
$$\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2 = 6.$$

[Note: $A_r (B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$

at two points A_r and B_r .]

Ans. 215

Sol.



$$(PA_1)(PB_1) = (PA_2)(PB_2) = \dots \dots (PA_5)(PB_5) = PT^2$$

$$PA_1^2 + PB_1^2 + 2(PA_1)(PB_1) = 25$$

$$PA_2^2 + PB_2^2 + 2(PA_2)(PB_2) = 36$$

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$$PA_5^2 + PB_5^2 + 2(PA_5)(PB_5) = 81$$

$$\text{add } \sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2 + 2 \sum_{i=1}^5 (PA_i)(PB_i) = 25 + 36 \dots + 81$$

$$\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2 = 25 + 36 + 49 + 64 + 81 - 2(S(PT))^2$$

$$= 255 - 10 \times 4$$

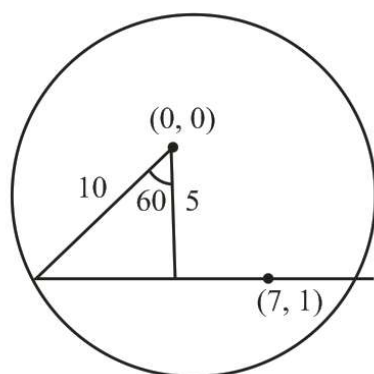
$$= 215$$

Q22. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point $(7, 1)$ and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of $m_1 m_2$, is

(A) -1 (B) 1 (C) $7/12$ (D) -3

Ans. A

Sol.



$$y - 1 = m(x - 7)$$

$$p = 5$$

$$\left(\frac{1 - 7m}{\sqrt{1 + m^2}} \right) = 5$$

$$49m^2 - 14m + 1 = 25 + 25m^2$$

$$24m^2 - 14m - 24 = 0$$

$$12m^2 - 7m - 12 = 0$$

$$12m^2 - -16m + 9m - 12 = 0$$

$$4m(3m - 4) + 3(3m - 4) = 0$$

$$m = \frac{-3}{4} \text{ or } m = \frac{4}{3}$$

$$m_1 m_2 = -1$$

Q23. Find the locus of the mid point of the chord of a circle $x^2 + y^2 = 4$ such that the segment intercepted by the chord on the curve $x^2 - 2x - 2y = 0$ subtends a right angle at the origin.

Ans. $x^2 + y^2 - 2x - 2y = 0$

Sol. Let the mid-point be $(h, k) \Rightarrow hx + ky = h^2 + k^2$

subtend right angle $\Rightarrow x^2 - 2(x - y)\left(\frac{hx + ky}{h^2 + k^2}\right) = 0 \Rightarrow (h^2 + k^2)x^2 - 2(x + y)(hx + ky) = 0$

As angle 90° , Coefficient of $x^2 +$ Coefficient of $y^2 = 0 \Rightarrow h^2 + k^2 - 2h - 2k = 0$

$\Rightarrow$ Locus $x^2 + y^2 - 2x - 2y = 0$

Q24. Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining $(0, 0)$ & the point of intersection of the chords with the circles make equal angle with axis of x.

Ans. $x + y = 2$

Sol. Let mid point $P(h, k)$

$T = S,$

$hx + ky - (h + x) - (k + y) = h^2 + k^2 - 2h - 2k$

$(h - 1)x + (k - 1)y = h^2 + k^2 - h - k$

$H - F = \frac{(h - 1) + (k - 1)y}{h^2 + k^2 - h - k}$

$x^2 = y^2 - 2x\left(\frac{(h - 1)x + (k - 1)y}{h^2 + k^2 - h - k}\right) - 2y\left(\frac{(h - 1)x + (k - 1)y}{h^2 + k^2 - h - k}\right) = 0$

Coeff of $xy = 0$

$\frac{-2(k - 1)}{h^2 + k^2 - h - k} \frac{-2(h - 1)}{h^2 + k^2 - h - k} = 0$

$k - 1 + h - 1 = 0$

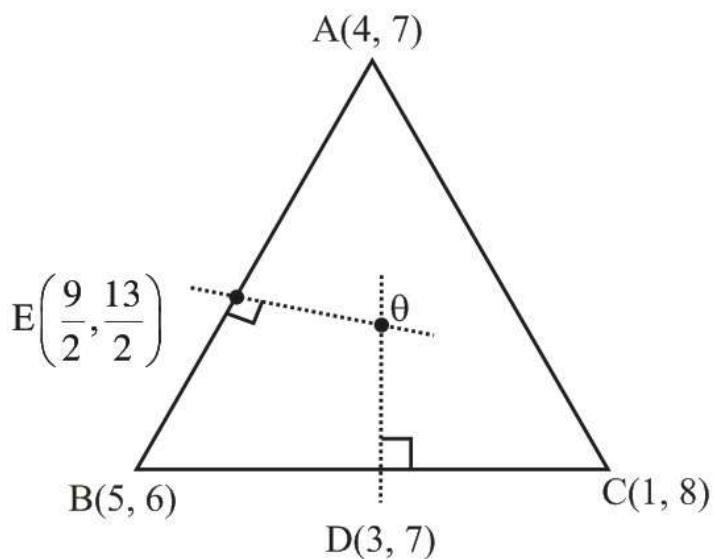
$h + k = 2$

$x + y = 2$

Q25. Find the equation of the circle passing through the three points $(4, 7)$, $(5, 6)$ and $(1, 8)$. Also find the coordinates of the point of intersection of the tangents to the circle at the points where it is cut by the straight line $5x + y + 17 = 0$.

Ans. $(-4, 2), x^2 + y^2 - 2x - 6y - 15 = 0$

Sol.



$$M_{OD} = 2$$

$$M_{OE} = 1$$

Equation of OD

$$y - 7 = 2(x - 3)$$

$$2x - y + 1 = 0 \quad \dots\dots\dots (i)$$

Equation of OE

$$y - \frac{13}{2} = 1\left(x - \frac{9}{2}\right)$$

$$2x - 2y + 4 = 0$$

$$x - y + 2 = \quad \dots\dots\dots (ii)$$

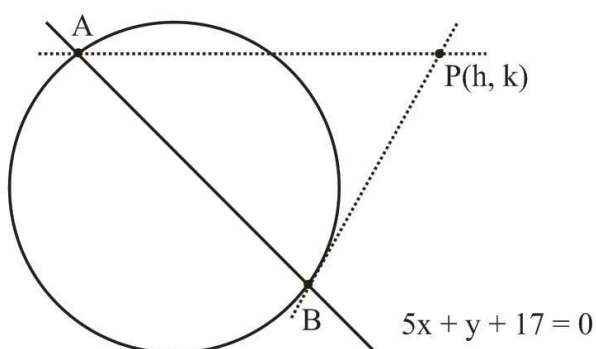
Center of circle is P.O.I. of $2x - y = 0$ and $x - y + 2 = 0$

$$O(1, 3)$$

$$r = 5$$

$$\text{equation of circle } (x - 1)^2 + (y - 3)^2 = 25$$

$$x^2 + y^2 - 2x - 6y - 15 = 0$$



Equation of AB $\Rightarrow T = 0$

$$hx + ky - (x + h) - 3(y + k) - 15 = 0$$

$$(h - 1) + (k - 3)y - h - 3k - 15 = 0$$

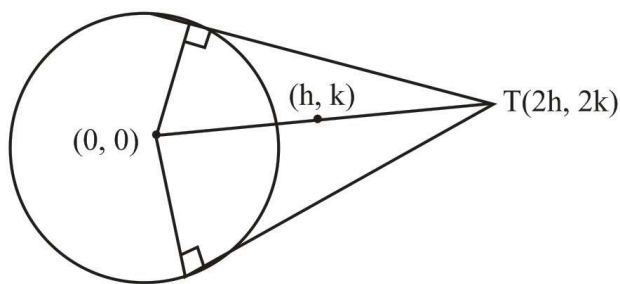
$$\frac{h - 1}{5} = \frac{k - 3}{1} = \frac{-h - 3k - 15}{17}$$

$$(-4, 2)$$

Q26. Tangents TP and TQ are drawn from a point T to the circle $x^2 + y^2 = a^2$. If the point T lies on the line $px + qy = r$, find the locus of centre of the circumcircle of triangle TPQ.

Ans. $2px + 2qy = r$

Sol.



T lies on the line $px + qy = r$

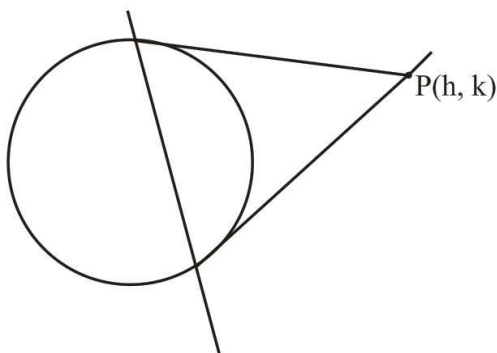
$$2ph + 2qk = r$$

$$rpx + 2qy = r$$

Q27. Find the point of intersection of the tangents drawn to the circle $x^2 + y^2 - 4x + 3y - 1 = 0$ at the points where the line $2x + y + 12 = 0$ meets it.

Ans. $(1, -2)$

Sol.



$$2x + y = 4 \quad \dots\dots (i)$$

equation of chord $\rightarrow T = 0$

$$hx + ky - 2(h + x) + \frac{3}{2}(k + y) - 1 = 0 \quad \dots\dots (ii)$$

compair (i) & (ii)

$$(h-2)x + \left(k + \frac{3}{2}\right)y - 2h + \frac{3k}{2} - 1 = 0$$

$$\frac{h-2}{2} = \frac{k + \frac{3}{2}}{1} = \frac{2h - \frac{3k}{2} + 1}{4}$$

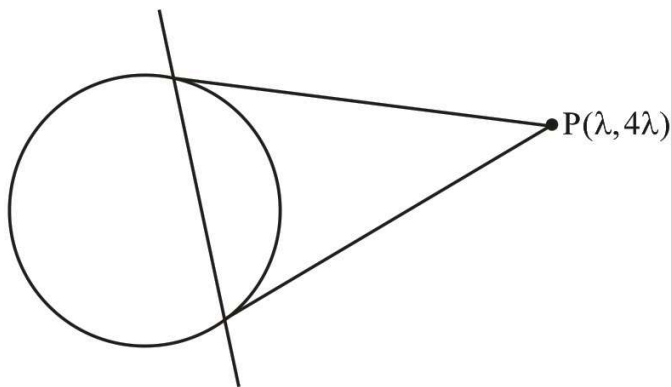
$$h = 1, k = -2$$

Q28. The chords of contact of the pair of tangents to the circle $x^2 + y^2 = 1$ drawn from any point on the line

$2x + y = 4$ pass through the point (a, b) , then find $\alpha_2 + \beta_2$.

Ans. $\frac{5}{16}$

Sol.



equation of C.O.C

$$T = 0$$

$$\lambda x + y(4 - 2\lambda) - 1 = 0$$

$$(4y - 1) + \lambda(x - 2y) = 0$$

$$L_1 : 4y - 1 = 0$$

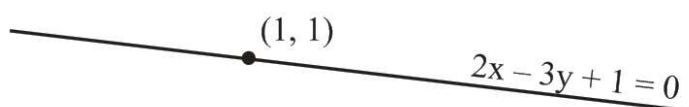
$$L_2 : x - 2y = 0$$

$$P.O.I. \left(\frac{1}{2}, \frac{1}{4}\right)$$

Q29. The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at $(1, 1)$. If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S .

Ans. $x^2 + y^2 - 6x + 4y = 0$ or $x^2 + y^2 + 2x - 8y + 4 = 0$

Sol.



$$S_p + \lambda L = 0$$

$$(x-1)^2 + (y-1)^2 + \lambda(2x-3y+1) = 0 \quad \dots\dots (i)$$

$$r = \sqrt{g^2 + f^2 - c} = \sqrt{13}$$

$$x^2 + y^2 - 2x + 2\lambda x - 2y - 3\lambda y + 2 + \lambda = 0$$

$$x^2 + y^2 + 2x(\lambda-1) - 2y\left(1 + \frac{3\lambda}{2}\right) + (2+\lambda) = 0$$

$$\sqrt{(\lambda-1)^2 + \left(1 + \frac{3\lambda}{2}\right)^2} - (2+\lambda) = \sqrt{13}$$

get λ and put in equation (i)

$$x^2 + y^2 - 6x + 4y = 0 \text{ and } x^2 + y^2 + 2x - 8y + 4 = 0$$

Q30. Find the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it.

Ans. $x^2 + y^2 + x - 6y + 3 = 0$

Sol. Equation of tangent to circle

$$x^2 + y^2 + 4x - 6y - 3 = 0 \text{ at } (2, 3) \Rightarrow 2x + 3y + 2(x+2) - 3(y+3) - 3 = 0$$

$$4x - 8 = 0 \quad \Rightarrow \quad x - 2 = 0$$

$$\text{family of circle} \Rightarrow S + \lambda L = 0$$

$$x^2 + y^2 + 4x - 6y - 3 + \lambda(x-2) = 0 \quad \dots\dots(y+3). (1)$$

Passes through (1, 1)

$$-3 - \lambda = 0 \Rightarrow \lambda = -3$$

Putting in (1)

$$x^2 + y^2 + x - 6y + 3 = 0$$

Q31. Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point (-1, -1).

Ans. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

Sol. equation of tangent at (-1, -1) on the circle

$$x^2 + y^2 - 4x - 6y - 12 = 0$$

$$T = 0$$

$$-x - y - 2(x-1) - 3(y-1) - 12 = 0$$

$$-3x - 4y - 7 = 0$$

$$3x + 4y + 7 = 0$$

$$(x+1)^2 + (y+1)^2 + \lambda(3x+4y+7) = 0$$

$$x^2 + y^2 = x(2 + 3\lambda) + 2y(1 + 2\lambda) + 2 + 7\lambda = 0$$

$$r = 3$$

$$\left(\frac{2 + 3\lambda}{2}\right) + (1 + 2\lambda) = 9$$

$$4 + 12\lambda + 9\lambda^2 + 16\lambda^2 + 16\lambda + 4 - 8 - 28\lambda = 36$$

$$25\lambda^2 = 36$$

$$\lambda = \pm \frac{6}{5}$$

$$\lambda = \frac{-6}{5}$$

or

$$\lambda = \frac{6}{5}$$

$$5x^2 + 5y^2 - 8x - 14y - 32 = 0$$

Riject

because circle not

touching intenally

Q32. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.

Ans. $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$

Sol. Equation of the any circle through the points of intersection of given circles is

$$(x^2 + y^2 - 2x - 4y - 4) + \lambda(x^2 + y^2 - 10x - 12y + 40) = 0$$

$$\Rightarrow x^2(1 + \lambda) + y^2(1 + \lambda) - 2x(1 + 5\lambda) - 2y(2 + 6\lambda) - 4 + 40\lambda = 0$$

$$\text{or } x^2 + y^2 - 2x \frac{(1 + 5\lambda)}{(1 + \lambda)} - 2y \frac{(2 + 6\lambda)}{(1 + \lambda)} + \frac{(40\lambda - 4)}{(1 + \lambda)} = 0 \quad \dots(1)$$

$$\text{Its radius } \sqrt{\left(\frac{1 + 5\lambda}{1 + \lambda}\right)^2 + \left(\frac{2 + 6\lambda}{1 + \lambda}\right)^2 - \left(\frac{40\lambda - 4}{1 + \lambda}\right)} = 4 \quad (\text{given})$$

$$\Rightarrow \frac{(1 + 5\lambda)^2 + (2 + 6\lambda)^2 - (40\lambda - 4)(1 + \lambda)}{(1 + \lambda)^2} = 16$$

$$\Rightarrow 5\lambda^2 + 34\lambda - 7 = 0$$

$$\text{or } (\lambda - 7)(5\lambda + 1) = 0$$

$$\therefore \lambda = 7 \text{ or } \lambda = \frac{1}{5}$$

Substituting the values of λ in (1), the required circles are

$$2x^2 + 2y^2 - 18x - 22y + 69 = 0 \text{ and } x^2 + y^2 - 2y - 15 = 0$$

Q33. Find the equation to the circle which passes through the points $(1, -2)$ and $(4, -3)$ and which has its centre on the straight line $3x + 4y = 7$.

Ans. $15x^2 + 15y^2 - 94x + 18y + 55 = 0$

Sol. $S_{AB} + \lambda L_{AB} = 0$

$$(x-1)(x-4) + (y+2)(y+3) + \lambda(x+3y+5) = 0$$

$$x^2 + y^2 - 5x + \lambda x + 5y + 3\lambda y + 10 + 5\lambda = 0$$

$$x^2 + y^2 - 2x\left(\frac{5}{2} - \frac{\lambda}{2}\right) - 2y\left(\frac{-5}{2} - \frac{3\lambda}{2}\right) + (10 + 5\lambda) = 0$$

Centere $\left(\frac{5}{2} - \frac{\lambda}{2}, \frac{-5}{2} - \frac{3\lambda}{2}\right)$ lies on the line $3x + 4y = 7$

$$3\left(\frac{5}{2} - \frac{\lambda}{2}\right) + \left(\frac{-5}{2} - \frac{3\lambda}{2}\right) = 7$$

get λ

$$15 - 3\lambda - 20 - 12\lambda = 14$$

$$-15\lambda = 19$$

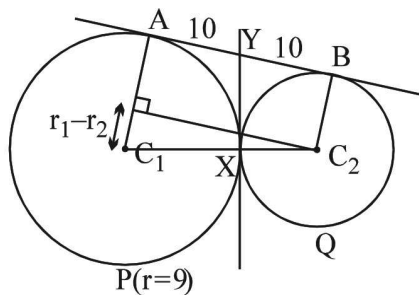
$$\lambda = \frac{-19}{15}$$

Q34. Let P and Q be 2 circles externally tangent at point X. A straight line is tangent to P at point A and is tangent to Q at point B ($A \neq B$). The line tangent to P and Q at X intersects line AB at a point Y.

If $AY = 10$ and the radius of P is 9. If the radius of the circle Q can be expressed as a rational number a/b in the lowest form, find the value of $(a + b)$.

Ans. 109

Sol.



$$AY = BY \quad (\because Y \text{ lies on radical axis})$$

$$(AB)^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2$$

$$AB = 2\sqrt{r_1r_2}$$

$$AB = 20 = 2\sqrt{9r_2}$$

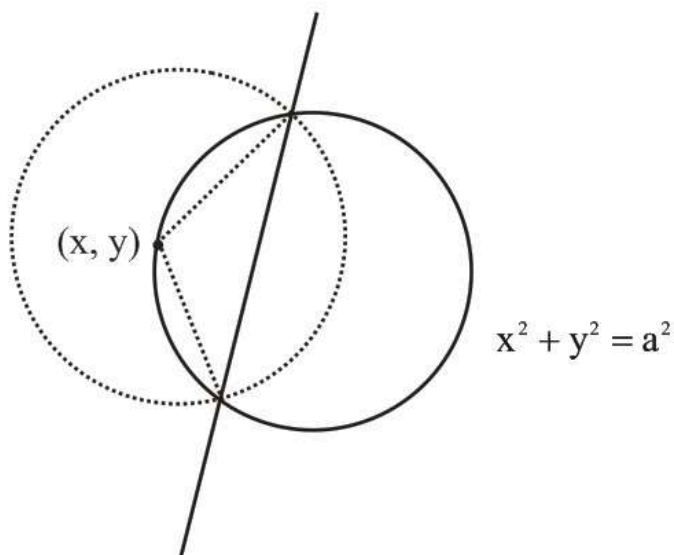
$$r_2 = \frac{100}{9} \text{ Ans.}$$

Q35. Find the sum of all integral value(s) of 'r' ($r > 0$) for which the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ intersect in two real distinct points.

Ans. 119

Sol. We have $(x-5)^2 + (y+8)^2 = 25 + 64 + r^2 - 89$

Sol.



equation of circle whose centre is (x, y) and radius is d .

$$(x - x_1)^2 + (y - y_1)^2 = d^2$$

Required line is common chord of circles.

So equation of Req. line is

$$S_1 - S_2 = 0$$

$$xx_1 + yy_1 - a^2 + \frac{d^2}{a} = 0$$

$$S_1 - S_2 = 0$$

$$6x + 14y + a + b = 0$$

passing through $(1, -4)$

$$a + b = 50$$

$$A.M. \leq G.M$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$ab \leq 625$$

Q38. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab) .

Ans. 625

Sol. -

Q39. Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ & $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.

Ans. $4x^2 + 4y^2 + 6x + 10y - 1 = 0$

Sol. equation of radical axis is :

$$S_1 - S_2 = 0$$

$$-5x + y - \frac{5}{2} = 0$$

$$-10x + 2y - 5 = 0$$

$$10x - 2y + 5 = 0$$

equation of required circle

$$S_1 + \lambda S_2 = 0$$

$$\left(x^2 + y^2 - x + 3y - \frac{3}{2}\right) + \left(x^2 + y^2 + 4x + 2y + 1\right) = 0$$

$$x^2 + y^2 + x\left(\frac{4\lambda - 1}{\lambda + 1}\right) + y\left(\frac{3 + 2\lambda}{1 + \lambda}\right) + \frac{\lambda - \frac{3}{2}}{\lambda + 1} = 0$$

$$\text{Centre} \left(-\frac{1}{2}\left(\frac{4\lambda - 1}{\lambda + 1}\right), -\frac{1}{2}\left(\frac{3 + 2\lambda}{\lambda + 1}\right) \right)$$

$$-5\left(\frac{4\lambda - 1}{\lambda + 1}\right) + \left(\frac{3 + 2\lambda}{\lambda + 1}\right) + 5 = 0$$

$$-20\lambda + 5 + 3 + 2\lambda + 5\lambda + 5 = 0$$

$$13\lambda = 13$$

$$\lambda = 1$$

Q40. Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ & cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Ans. $x^2 + y^2 + 16x + 14y - 12 = 0$

Sol. $S_1 + \lambda S_2 = 0$

$$\left(x^2 + y^2 - x - 6y - 12\right) + \lambda\left(x^2 + y^2 - x - 6x + 4y - 12\right) = 0$$

$$x^2 + y^2 + 2x\left(\frac{3\lambda - 2}{\lambda + 1}\right) + 2y\left(\frac{2\lambda - 3}{\lambda + 1}\right) - \left(\frac{12 + 12\lambda}{\lambda + 1}\right) = 0$$

$$x^2 + y^2 - 2x - 4 = 0$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$2(-1)\left(\frac{3\lambda - 2}{\lambda + 1}\right) + 0 = \frac{-12 - 12\lambda}{\lambda + 1} - 4$$

$$\frac{6\lambda - 4}{\lambda + 1} = \frac{12 + 12\lambda + 4\lambda + 4}{\lambda + 1}$$

$$10\lambda = -20$$

$$\lambda = -2$$

Q41. The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ & $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.

Ans. $(-4, 4)$; $(-1/2, 1/2)$

Sol. Let equation of circle

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Centre $(-g, -f)$ lies on the line $2x - 2y + 9 = 0$

$$-2g + 2f + 9 = 0 \quad \dots (1)$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$0 + 0 = c - 4$$

$$c = 4$$

$$x^2 + y^2 = 2x\left(\frac{2f+9}{2}\right) + 2fg + 4 = 0$$

$$x^2 + y^2 + 9x + 4 + f(2x + 2y) = 0$$

$$S : x^2 + y^2 + 9x + 4 = 0$$

$$L : x + y = 0$$

$$\text{P.O.I. } (-4, 4) \text{ and } \left(-\frac{1}{2}, \frac{1}{2}\right)$$

Q42. Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k .

Ans. $x^2 + y^2 + 4x - 6y = 0$; $k = 1$

Sol. Let the equation of circle be

$$x^2 + y^2 + 2gx + 2fy = 0$$

it passes through origin)

The line pair is

$$xy - 3x + 2y - 6 = 0$$

$$x(y - 3) + 2(y - 3) = 0$$

$$(x + 2)(y - 3) = 0$$

The centre is point of intersection of these two lines.

$$c \equiv (-2, 3)$$

$$g = 2 \text{ and } f = -3$$

$$s \equiv x^2 + y^2 + 4x - 6y = 0$$

$$s_1 = x^2 + y^2 - kx + 2ky - 8 = 0$$

Since s & s_1 are orthogonal

$$\therefore 2gg_1 + 2ff_1 = 0 - 8$$

$$\Rightarrow 2(-k) + (-3) \times 2k = 0 - 8$$

$$\therefore k = 1$$

Q43. Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.

Ans. $x^2 + y^2 = 64$

Sol. $x^2 + y^2 - 14x - 8y + 64 = 0$

$$(g, f) = (7, 4); c_1 = 64$$

$$\text{Let } x^2 + y^2 = a^2$$

$$\Rightarrow x^2 + y^2 - a^2 = 0$$

$$(g, f) = (0, 0); c_2 = -a^2$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$0 + 0 = 64 + (-a^2)$$

$$a^2 = 64$$

$$\text{i.e., } x^2 + y^2 = 64$$

Q44. Find the equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally.

Ans. $x^2 + y^2 + 7x - 11y + 38 = 0$

Sol. $S_p + \lambda L = 0$

$$(x + y)^2 + (y - 7)^2 + \lambda(x + 4) + y(\lambda + 4) + y(\lambda - 14) + 53 - 5\lambda = 0$$

$$2\left(\frac{\lambda + 4}{2}\right)(2) + \left(\frac{\lambda - 14}{2}\right)(-3) = 9 + 53 - 5\lambda$$

$$2\lambda + 8 + 42 - 3\lambda = 9 + 53 - 5\lambda$$

$$4\lambda = 12$$

$$\lambda = 3$$

Q45. Find the equation of the circle passing through the point $(-6, 0)$ if the power of the point $(1, 1)$ w.r.t. the circle is 5 and it cuts the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ orthogonally.

Ans. $x^2 + y^2 + 6x - 3y = 0$

Sol. $x^2 + y^2 + 2gx + 2fy + c = 0$

$$S_1 = 5$$

$$1 + 1 + 2g + 2f + c = 5$$

$$2g = 2f + c = 3 \quad \dots (1)$$

passing through $(-6, 0)$

$$36 - 12g + c = 0 \quad \dots (2)$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$2g(-2) + 2f(-3) = c - 3$$

$$-4g - 6f = c - 3 \quad \dots (3)$$

get g , f and c

$$g = \frac{-5}{2}, f = \frac{-5}{4}, c = \frac{1}{2}$$

- Q46. (a) Find the equation of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circle $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally.

Ans. (a) $x^2 + y^2 + 14x - 6y + 6 = 0$;

Sol. $S_1 + \lambda S_2 = 0$

$$(x^2 + y^2 - 2x - 6y + 6) + \lambda(x^2 + y^2 + 2x - 6y + 6) = 0$$

$$x^2 + y^2 + 2x\left(\frac{\lambda - 1}{\lambda + 1}\right) - 6y\left(\frac{\lambda + 1}{\lambda + 1}\right) + \frac{6(\lambda + 1)}{\lambda + 1} = 0$$

$$2\left(\frac{\lambda - 1}{\lambda + 1}\right)(2) + 2(-3)(3) = 6 + 4$$

$$\lambda - 1 = 7\lambda + 7 \Rightarrow \lambda = \frac{-4}{3}$$

- Q47. Line $2x + 3y + 1 = 0$ is a tangent to a circle at $(1, -1)$. This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points $(0, -1)$ and $(-2, 3)$. Find equation of circle.

Ans. $2x^2 + 2y^2 - 10x - 5y + 1 = 0$

Sol. $(x - 1)^2 + (y + 1)^2 + \lambda(2x + 3y + 1) = 0$

$$x^2 + y^2 + 2x(\lambda - 1) + 2y\left(1 + \frac{3\lambda}{2}\right) + 2 + \lambda = 0 \quad \dots (1)$$

$$(x)(x + 2) + (y + 1)(y - 3) = 0$$

$$x^2 = y^2 + 2x - 2y - 3 = 0 \quad \dots (2)$$

$$2(\lambda - 1)(1) + 2\left(1 + \frac{3\lambda}{2}\right)(-1) = -3 + 2 + \lambda$$

$$2\lambda - 2 - 2 - 3\lambda = -3 + 2 + \lambda$$

$$-\lambda - 4 = -1 + \lambda$$

$$\lambda = \frac{-3}{2}$$

Paragraph

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$, k being a constant, then the line L always touches a fixed circle C.

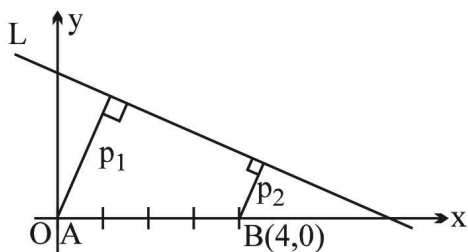
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Q48. The centre of the circle C lies on

- (A) line segment joining AB (B) perpendicular bisector of AB (C) one of A or B
(D) nothing definite can be said

Ans. A

Sol.



Let A = (0, 0) and B = (4, 0) and the line be $ax + by = 1$

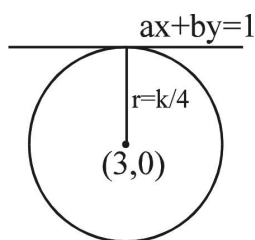
$$P_1 = \left| \frac{1}{\sqrt{a^2 + b^2}} \right|; P_2 = \left| \frac{4a - 1}{\sqrt{a^2 + b^2}} \right|$$

$$p_1 + 3p_2 = k$$

$$\left| \frac{1}{\sqrt{a^2 + b^2}} \right| + 3 \left| \frac{4a - 1}{\sqrt{a^2 + b^2}} \right| = k$$

now (0, 0) and (4, 0) must give the same sign i.e. - ve with the line L ($4a - 1 < 0$)

$$\therefore \frac{1}{\sqrt{a^2 + b^2}} + \frac{3(1 - 4a)}{\sqrt{a^2 + b^2}} = k$$



$$\left| \frac{4(1 - 3a)}{\sqrt{a^2 + b^2}} \right| = k; \left| \frac{3a - 1}{\sqrt{a^2 + b^2}} \right| = \frac{k}{4}$$

hence centre of the fixed circle is (3, 0) which lies on the line segment AB $\Rightarrow$ (A)

Paragraph

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$, k being a constant, then the line L always touches a fixed circle C.

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Q49. If $k = 4$ then the radius of the circle is

- (A) 1 (B) 2 (C) 4 (D) 8

Ans. A

Sol. If $k = 4 \Rightarrow r = 1$

Paragraph

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$, k being a constant, then the line L always touches a fixed circle C.

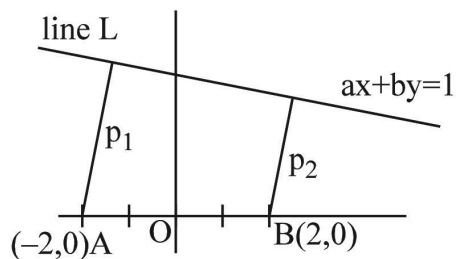
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Q50. If A and B are $(-2, 0)$ and $(2, 0)$ respectively, then the centre of the circle C is

- (A) $(0, 1)$ (B) $(1, 0)$ (C) $(\frac{3}{2}, 0)$ (D) can not be found

Ans. B

Sol.



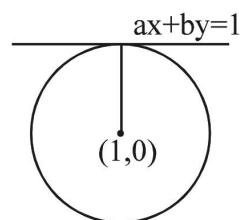
$$P_1 = \left| \frac{-2a - 1}{\sqrt{a^2 + b^2}} \right|; P_2 = \left| \frac{2a - 1}{\sqrt{a^2 + b^2}} \right|$$

hence, with the same argument, $2a - 1 < 0$ and $-2a - 1 < 0$

$$\therefore p_1 + 3p_2 = k$$

$$\frac{1 + 2a}{\sqrt{a^2 + b^2}} + \frac{3(1 - 2a)}{\sqrt{a^2 + b^2}} = k$$

$$\frac{4(1 - a)}{\sqrt{a^2 + b^2}} = k$$



$$\frac{|a-1|}{\sqrt{a^2+b^2}} = \frac{k}{4}$$

hence centre is (1, 0) **Ans.**

Paragraph

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

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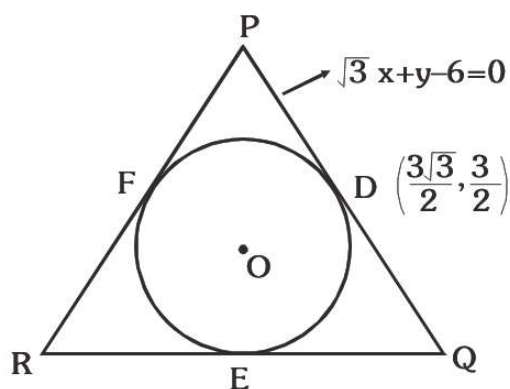
Q51. The equation of circle C is

(A) $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$ (B) $(x - 2\sqrt{3})^2 + (y + \frac{1}{2})^2 = 1$

(C) $(x - \sqrt{3})^2 + (y + 1)^2 = 1$ (D) $(x - \sqrt{3})^2 + (y - 1)^2 = 1$

Ans. D

Sol.



$$m_{PQ} = -\sqrt{3}$$

$$\text{so slope of OD} = \frac{1}{\sqrt{3}}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \frac{x - \frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = \frac{y - \frac{3}{2}}{\frac{1}{2}} = 1$$

$$(2\sqrt{3}, 2) \text{ (not possible) \& } (\sqrt{3}, 1)$$

$$\text{hence circle is } (x - \sqrt{3})^2 + (y - 1)^2 = 1$$

Paragraph

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$

$x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

\n

Q52. Points E and F are given by

- (A) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\sqrt{3}, 0\right)$ (B) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \left(\sqrt{3}, 0\right)$ (C) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
 (D) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Ans. A

Sol. For point E $\frac{x - \sqrt{3}}{-\frac{3}{2}} = \frac{y - 1}{\frac{1}{2}} = 1 \quad \left[\because E\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right) \right]$

For point F $\frac{x - \sqrt{3}}{0} = \frac{y - 1}{-1} = 1 \quad \left[\because F(\sqrt{3}, 0) \right]$

Paragraph

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

\n

Q53. Equations of the sides RP, RQ are

- (A) $y = \frac{2}{\sqrt{3}}x + 1, y = -\frac{2}{\sqrt{3}}x - 1$ (B) $y = \frac{1}{\sqrt{3}}x, y = 0$ (C) $y = \frac{\sqrt{3}}{2}x + 1, y = -\frac{\sqrt{3}}{2}x - 1$
 (D) $y = \sqrt{3}x, y = 0$

Ans. D

Sol. Equation of line RP $y = 0$

Equation of line QR $y - \frac{3}{2} = \sqrt{3}\left(x - \frac{\sqrt{3}}{2}\right)$

$y = \sqrt{3}x$

Paragraph

Let S_1, S_2 and S_3 be the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ and $x^2 + y^2 - 4y - 5 = 0$ respectively, then

\n

Q54. Point (2, 2) lies

- (A) in the circle S_1 only (B) in the circle S_2 only (C) in the circle S_3 only

(D) in the circles S_1 and S_2 but not in S_3

Ans. C

Sol. $S_3 = 4 + 4 - 8 - 5$

$$S_3 < 0$$

in the circle S_3

Paragraph

Let S_1 , S_2 and S_3 be the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ and $x^2 + y^2 - 4y - 5 = 0$ respectively, then

\n

Q55. The point $(1, 2)$ lies

(A) in circles S_2 and S_3 but not in circle S_1

(B) in circles S_2 and S_1 but not in circle S_3

(C) in circle S_3 but not in circles S_1 and S_2 (D) in circles S_1 , S_2 and S_3

Ans. C

Sol. $\Rightarrow S_3 = 1 + 4 - 8 - 5$

$$S_3 < 0$$

in the circle S_3

$$\Rightarrow S_1 = 1 + 4 - 4$$

$$S_1 > 0$$

not in circle S_1

$$\Rightarrow S_2 = 1 + 4 - 6 + 5$$

$$S_2 > 0$$

not in circle S_2

Paragraph

Let S_1 , S_2 and S_3 be the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 6x + 5 = 0$ and $x^2 + y^2 - 4y - 5 = 0$ respectively, then

\n

Q56. The line $x - y + 2\sqrt{2} = 0$ is

(A) tangent to S_1 and does not meet S_2 and S_3

(B) tangent to S_1 , secant of S_3 and does not meet S_2

- (C) tangent to S_1 and secant of S_2 and S_3 (D) tangent to S_3 and secant of S_1 and S_2

Ans. B

Sol. For S_1

$$p = \left| \frac{2\sqrt{2}}{\sqrt{2}} \right| = 2$$

$$p = r$$

tangent to S_1

for S_2

$$p = \left| \frac{3 + 2\sqrt{2}}{\sqrt{2}} \right|, \quad r = 2$$

$$p > r$$

does not meet S_2

for S_3

$$p = \left| \frac{-2 + 2\sqrt{2}}{\sqrt{2}} \right|, \quad r = 3$$

$$p < r$$

Secant for S_3

Q57. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. The point(s) on this circle, the tangent(s) at which is/are parallel to the straight line joining the origin to its centre is/are :

- (A) $(1, -5)$ (B) $(5, 1)$ (C) $(-5, -1)$ (D) $(-1, 5)$

Ans. D

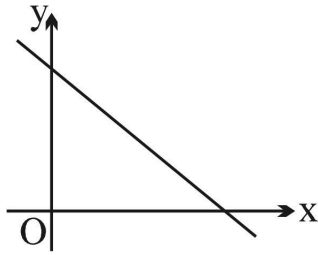
Sol. Note that D is right angled at $(0, 6)$. Centre of the circle is $(2, 3)$. Slope of the line joining origin to the centre is $\frac{3}{2}$. Take parametric equation of a line through $(2, 3)$ with $\tan\theta = -\frac{2}{3}$ as $\frac{x-2}{\cos\theta} = \frac{y-3}{\sin\theta} = \pm r$ where $r = \sqrt{13}$. Get the co-ordinates on the circle.

Q58. Three distinct lines are drawn in a plane. Suppose there exist exactly n circles in the plane tangent to all the three lines, then the possible values of n is/are

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. A, C, D

Sol.



Case-1: If lines form a triangle then $n = 4$

i.e. 3 excircles and 1 incircle

Case-2: If lines are concurrent or all 3 parallel then $n = 0$

Case-3: If two are parallel and third cuts then $n = 2$

hence (A), (C), (D).

Q59. Equations of circles which pass through the points $(1, -2)$ and $(3, -4)$ and touch the x-axis is

(A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$

(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) none

Ans. C, B

Sol. Let equation of required circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

it passes through $(1, -2)$ & $(3, -4)$

$$2g - 4f + c = -5$$

$$6g - 8f + c = -25$$

$$4g - 8f + 2c = -10$$

$$6g - 8f + c = -25$$

$$-2g + c = 15$$

circle touches x-axis $g^2 = c \Rightarrow$

$$g^2 - 2g - 15 = 0$$

$$g = 5, -3$$

$$g = 5, \quad c = 25, \quad f = 10 \Rightarrow x^2 + y^2 + 10x + 20y + 25 = 0$$

$$g = -3, \quad c = 9, \quad f = 2 \Rightarrow x^2 + y^2 - 6x + 4y + 9 = 0$$

Q60. The circle $x^2 + y^2 - 2x - 3ky - 2 = 0$ passes through two fixed points,

(A) $(1 + \sqrt{3}, 0)$ (B) $(-1 + \sqrt{3}, 0)$ (C) $(-\sqrt{3}, -1, 0)$ (D) $(1 - \sqrt{3}, 0)$

Ans. A

Sol. Two fixed pts. are point of intersection of

$$x^2 + y^2 - 2x - 2 = 0 \quad \& \quad y = 0$$

Point $x^2 - 2x - 2 = 0$

$$(x - 1)^2 - 3 = 0 \Rightarrow x - 1 = \sqrt{3}, x - 1 = -\sqrt{3} \Rightarrow (1 + \sqrt{3}, 0) (1 - \sqrt{3}, 0)$$

Q61. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is $(x - c)^2 + (y - c)^2 = c^2$ where c is

- (A) 1 (B) 2 (C) 4 (D) 6

Ans. D, A

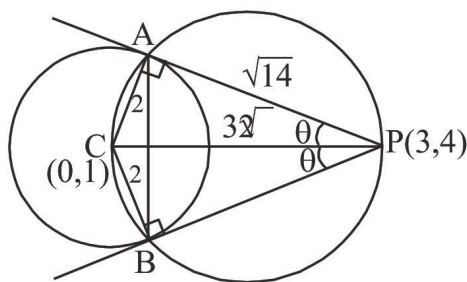
Sol. $\left| \frac{4C + 3C - 12}{5} \right| = C \Rightarrow C = 1, 6$

Q62. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point $P(3, 4)$. Which of the following alternative(s) is/are correct?

- (A) The power of point $P(3, 4)$ with respect to circle $S = 0$ is 14.
 (B) The angle between tangents from $P(3, 4)$ to the circle $S = 0$ is $\frac{\pi}{3}$
 (C) The equation of circumcircle of ΔPAB is $x^2 + y^2 - 3x - 5y + 4 = 0$
 (D) The area of quadrilateral PACB is $3\sqrt{7}$ square units where C is the centre of circle $S = 0$.

Ans. A, C

Sol.



Clearly power of point $P(3, 4)$ w.r.t. $S = 0$ is

$$S_1 = 14 \Rightarrow \text{option (A) is correct}$$

$$\text{Also } \sin \theta = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$$

$$\text{Now angle between tangents} = 2\sin^{-1} \frac{\sqrt{2}}{3}$$

Also CP is diameter of circumcircle of ΔPAB , so equation of circumcircle is

$$x(x - 3) + (y - 1)(y - 4) = 0 \Rightarrow x^2 + y^2 - 3x - 5y + 4 = 0 \Rightarrow \text{option (C) is correct}$$

Now, area of quadrilateral PACB $= 2 \times \frac{1}{2} \times 2 \times \sqrt{14} = 2\sqrt{14}$ square roots $\Rightarrow$ Option (D) is correct.

Q63. If the equation of circle touching the y-axis at (0, 3) and making an intercept of 8 unit on x-axis is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $(g + f + c)$ can be

- (A) 1 (B) 7 (C) 11 (D) 14

Ans. A, C

Sol. The circle is $x^2 + (y - 3)^2 + \lambda x = 0$

putting $y = 0$, we get $x^2 + \lambda x + 9 = 0$

x-intercept is $|x_1 - x_2|$

$$|x_1 - x_2|^2 = (x_1 + x_2)^2 - 4x_1x_2$$

$$64 = \lambda^2 - 36 \quad \lambda \Rightarrow \lambda^2 = 100 \Rightarrow \lambda = 10 \text{ or } -10$$

Solving $\lambda = \pm 10$

The circles are $x^2 + y^2 \pm 10x - 6y + 9 = 0$

$$\Rightarrow g = \pm 5; f = -3; c = 9 \Rightarrow (g + f + c) = 11 \text{ and } 1 \text{ Ans.}$$

Q64. The circles $x^2 + y^2 - 2x - 4y + 1 = 0$ and $x^2 + y^2 + 4x + 4y - 1 = 0$

- (A) touch internally (B) touch externally
(C) have $3x + 4y - 1 = 0$ as the common tangent at the point of contact.
(D) have $3x + 4y + 1 = 0$ as the common tangent at the point of contact.

Ans. B, C

Sol. -

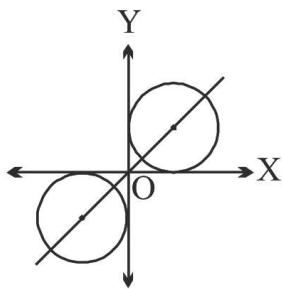
Q65. Which of the following is/are True?

The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that

- (A) they do not intersect. (B) they touch each other.
(C) their exterior common tangents are parallel.
(D) their interior common tangents are perpendicular.

Ans. A, C, D

Sol. Centre (3, 3) with radius = 3 and centre (-3, 3) with radius = 3.



Q66. Consider the circles $C_1: x^2 + y^2 = 16$ and $C_2: x^2 + y^2 - 12x + 32 = 0$. Which of the following statements is/are correct?

- (A) Number of common tangent to these circles is 3.
- (B) The point P with coordinates (4, 1) lies outside the circle C_1 and inside the circle C_2 .
- (C) Their direct common tangent intersect at (12, 0).
- (D) Slope of their radical axis is not defined.

Ans. A, C, D

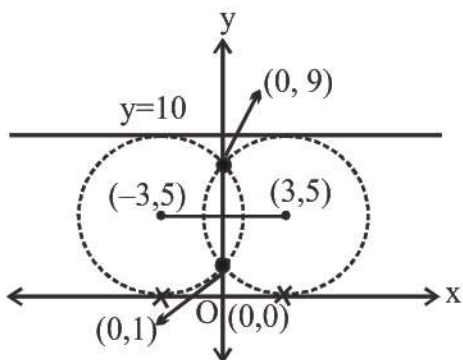
Sol. $C_1: (0, 0)$, radius 4
 $C_2: (6, 0)$, radius 2
 circle touch externally.

Q67. If a circle passes through $P(0, 1)$, $Q(0, 9)$ and touches the x – axis, then which of the following statement(s) is/are true?

- (A) Centres of circles are $(\pm 3, \pm 5)$.
- (B) Equation of one of their direct common tangent is $y = 10$.
- (C) Radii of both the circles is 5. (D) Length of common chord of circles is 8.

Ans. B, C, D

Sol.



Equation of any circle through $P(0, 1)$ and $Q(0, 9)$ is $x^2 + (y - 1)(y - 9) + \lambda x = 0$
 $\Rightarrow x^2 + y^2 - 10y + 9 + \lambda x = 0$

As above circle touches x - axis, so put $y = 0$

$$\Rightarrow x^2 + \lambda x + 9 = 0$$

should have equal roots. So, Discriminant = 0

$$\Rightarrow \lambda^2 = 36 \quad \therefore \lambda x + \pm 6$$

Hence required circles are $x^2 + y^2 + 6x - 10y + 9 = 0$

$$\text{and } x^2 + y^2 - 6x - 10y + 9 = 0$$

Q68. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in \mathbb{R}$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to

- (A) $-3/4$ (B) 0 (C) $3/4$ (D) 1

Ans. A, B

Sol. Line: $y = 1 + c(x + 3)$ is tangent to circle $x^2 + y^2 =$

$$\therefore p = r$$

$$\therefore \left| \frac{1 + 3c}{\sqrt{1 + c^2}} \right|$$

$$(1 + 3c)^2 = 1 + c^2$$

$$9c^2 + 6c = c^2$$

$$8c^2 = -6c$$

$$\therefore c = 0 \quad \text{or} \quad c = -\frac{3}{4} \Rightarrow (A), (B)$$

Q69. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?

- (A) $3x - y = 0$ (B) $x + 3y = 0$ (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$

Ans. A, B, C, D

Sol. Chords equidistance from the centre are equal

Q70. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :

- (A) $x = 0$ (B) $y = 0$ (C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

Ans. A, C

Sol. Let equation of tangent $y = mx$
apply C.O.T.

$$\left| \frac{mr - h}{\sqrt{r^2 + h^2}} \right| = r$$

$$m^2 r^2 + h^2 - 2mhr = m^2 + r^2$$

$$m \rightarrow \infty \quad \text{and} \quad m = \frac{b^2 - r^2}{2hr}$$

$$x = 0 \quad \text{and} \quad y = \left(\frac{h^2 - r^2}{2hr} \right) x$$

Q71. Match the column type

Column – I

Column – II

- | | |
|---|-------|
| (A) Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is | (P) 4 |
| (B) A chord of the circle $(x - 1)^2 + y^2 = 4$ lies along the line $y = 22\sqrt{3}(x - 1)$. The length of the chord is equal to | (Q) 2 |
| (C) The number of circles touching all the three lines $3x + 7y = 2$, $21x + 49y = 5$ and $9x + 21y = 0$ are | (R) 0 |
| (D) If radii of the smallest and largest circle passing through the point $(\sqrt{3}, \sqrt{2})$ and touching the circle $x^2 + y^2 - 2y - 2 = 0$ are r_1 and r_2 respectively, then the mean of r_1 and r_2 is | (S) 1 |

(A) $(A) \rightarrow Q, (B) \rightarrow P, (C) \rightarrow R, (D) \rightarrow S$

(B) $(A) \rightarrow P, (B) \rightarrow Q, (C) \rightarrow S, (D) \rightarrow R$

(C) $(A) \rightarrow S, (B) \rightarrow R, (C) \rightarrow P, (D) \rightarrow Q$

(D) $(A) \rightarrow R, (B) \rightarrow S, (C) \rightarrow Q, (D) \rightarrow P$

Ans. 5f041d4d900175a12c9ffc72

Sol. (A) $S_1 - S_2 = 0$ is the required common chord i.e. $2x = a$

Make homogeneous, we get $x^2 + y^2 - 8.4 \frac{x^2}{a^2} = 0$

As pair of lines subtending angle of 90° at origin

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

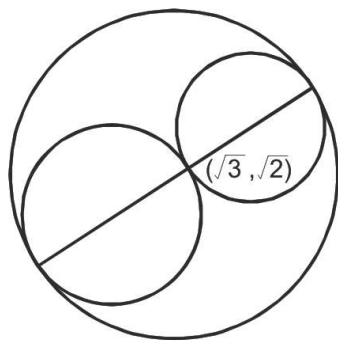
$\therefore a = \pm 4$

(B) $y = 22\sqrt{3}(x - 1)$ passes through centre $(1, 0)$ of circle

(C) Three lines are parallel so not any circle is possible

(D) $2(r_1 + r_2) = 4$

$$r_1 + r_2 \Rightarrow \frac{r_1 + r_2}{2} = 1$$



Q72.

Column – I

Column – II

(A) Number of common tangents of the circles

(P) 1

$$x^2 + y^2 - 2x = 0 \text{ and } x^2 + y^2 + 6x - 6y + 2 = 0 \text{ is}$$

(B) Number of transverse common tangents of the circles

(Q) 2

$$x^2 + y^2 - 4x - 10y + 4 = 0 \text{ \& } x^2 + y^2 - 6x - 12y - 55 = 0 \text{ is}$$

(C) Number of common tangents of the circles $x^2 + y^2 - 2x - 4y = 0$
& $x^2 + y^2 - 8y - 4 = 0$ is

(R) 3

(D) Number of direct common tangents of the circles

(S) 0

$$x^2 + y^2 + 2x - 8y + 13 = 0 \text{ \& } x^2 + y^2 - 6x - 2y + 6 = 0 \text{ is}$$

(A) (A) $\rightarrow$ R, (B) $\rightarrow$ S, (C) $\rightarrow$ P, (D) $\rightarrow$ Q

(B) (A) $\rightarrow$ S, (B) $\rightarrow$ R, (C) $\rightarrow$ P, (D) $\rightarrow$ Q

(C) (A) $\rightarrow$ R, (B) $\rightarrow$ S, (C) $\rightarrow$ Q, (D) $\rightarrow$ P

(D) (A) $\rightarrow$ Q, (B) $\rightarrow$ P, (C) $\rightarrow$ S, (D) $\rightarrow$ R

Ans. 5f041f9c90e4b0a1523a618a

Sol. (A) $C_1 (1, 0)$, $r_1 = 1$ and $C_2 (-3, 3)$, $r_2 = 4$
distance between centres C_1 and $C_2 = d = 5$

$$d = r_1 + r_2 = 5 \Rightarrow 3 \text{ common tangents}$$

(B) $C_1 (2, 5)$, $r_1 = 5$ and $C_2 (3, 6)$, $r_2 = 10$

$$\text{distance between centres } C_1 \text{ and } C_2 = d = \sqrt{2}$$

$$d < |r_1 - r_2|$$

$\Rightarrow$ no common tangent

(C) $C_1 (1, 2)$, $r_1 = \sqrt{5}$ and $C_2 (0, 4)$, $r_2 = 2\sqrt{5}$

$$\text{distance between centres } C_1 \text{ and } C_2 = d = |r_1 - r_2| = d$$

number of common tangents is 1

(D) $C_1 (-1, 4)$, $r_1 = 2$ and $C_2 (3, 1)$, $r_2 = 2$

$$\text{distance between centres } C_1 \text{ and } C_2 = d = 5$$

$$d > r_1 + r_2$$

$\Rightarrow$ number of direct common tangents is 2